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**Evolution in Gendered Language and Reflection of Feminist
Waves in Public Sphere between the 19th and 20th Centuries:
A Digital Analysis of Women's Status in Society**

A Thesis Submitted for Partial Fulfillment of the Requirements for the Degree of Master in
Language Technology and Digital Humanities

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Abstract

This thesis explores the subject of evolution in gendered language and reflection of feminist waves in the public sphere to disclose women's status in society from the 19th to the 20th century, the struggles they had to experience to secure their position in society. To investigate these, a few digital tools, such as some methods from Python, Voyant tools, and SketchEngine, were used. The output of Sentiment Analysis allowed us to identify the emotional tone of the documents and supporters' approaches related to the events. Then, Word Cloud of each text were printed to have insights about the key themes, and Pie Charts showed the main goals of each movement. The Named Entity Recognition task was performed to see the participants, including supporters and dissidents. Voyant tools such as Corpus Collocates, Mandala, Document Terms, and Collocates Graph provided the key terms together with their collocates, which helped to make connections between events. The Word Sketch technique from SketchEngine described the thematic term within the expressions in which they were used, thus we were able to see small but essential informative parts from each file. Lastly, and of equal importance, the Corpus of Historical American English was used to see the statistics of the most used terms during the first and second waves of feminism. During the research, the digital tools helped us understand the main issues of that period. In conclusion, the research contributes to the field of history by employing digital tools, showing that history can be learned more easily and efficiently with the help of technological possibilities.

Introduction. Historical Research and NLP

1. History and Technology

People work hard to form their identity in a way that they plan to see themselves in the future by analyzing today's agenda and perspectives, which constantly improve. In that sense, historians are reluctant to acknowledge the growing role of technology in the progression of history. As seen in the book "Consuming History" by Jerome de Groot (Crymble 2021, 2), books have already become outdated, as technological sources gradually replace them. Instead of focusing on discussions of technology, historians prefer writing philosophical works, such as those on objectivity vs. subjectivity, modernism vs. postmodernism, and so on. According to Ludmilla Jordanova, a more professional approach to technology may prevent the disappearance of history as a discipline (Crymble 2021, 2). She suggests that historians should work on new philosophical writings based on old knowledge, and these should be shaped using technological tools. To achieve this, new approaches and perspectives should emerge. It is an undeniable fact that historians should play a more active role in public affairs and participate in public policy. The more we use computers in our daily work, the more they become an indispensable part of our routine jobs, thus they can have a significant impact on our cognitive objectives as well. Historical and philosophical collections are preserved in different forms, such as books, articles, and blogs, giving rise to the term "digital history" to emerge. The historians still prefer to focus on theoretical discussion and ignore analyzing the capabilities of technological tools and how they transform historical works. Unfortunately, they refuse to acknowledge the new phase. Throughout the years, historians have attempted to describe the connection and interaction between history and technology with a few works, such as *The Historian and the Computer* (1971) and *Exploring Big Historical Data* (2016) (Crymble 2021, 2).

However, those works became a flop because the authors focused on the future and did not analyze the historical roots. Chad Black and Bethany Nowviskie state that they failed because they turned a blind eye to the past and did not even look through the period before the 1980s (Crymble 2021, 3). Historians worked on current practical issues and neglected to study the literature of their field. Classic works such as *Capitalism and Slavery* by Eric Williams in 1944 and *The Making of the English Working Class* by E.P. Thompson were still well-known even decades after they were published, containing their intellectual value (Crymble 2021, 3).

When questions about the results or benefits of digital studies were asked, historians would always say that they plan to see the advantages in the future. Cameron Blevins, a historian in the 19th century in the U.S., once said that historians' digital achievements were always future-oriented, instead of analyzing what contributions had been made in the past (Crymble 2021, 4). People were looking forward to seeing what would be achieved in the future rather than valuing what they had already been doing. In the past, some materials would be used just for doctoral dissertations, now those are studied by undergraduate students, which technological advancements make possible. However, the publication of documents that contain important information related to the past was stopped; therefore, misunderstandings and divergences emerged. The first journal ceased in 2004, which was called *Computers and the Humanities*, and the last one, called *Journal of Digital Humanities*, was in 2014 (Crymble 2021, 4). With the arrival of new journal repositories, historians changed their approach to previously analyzed research. Famous platforms such as Google Scholar and JSTOR became useful for scholars to share their research and historical texts; however, the forgotten historical texts were not uploaded to digital repositories since they had no advocates anymore (Crymble 2021, 4). After the closures of the journals, digital platforms such as blogs and social media were the ideal places to discuss computational scholarship. After a while, historians focused on the current situation of the digital sphere rather than searching for its origin, and they attempted to find a proper definition for e-scholarship. Deborah Lines Anderson, who explained the current situation of digital scholarship as non-stop instability, made the first effort in 2002 (Crymble 2021, 5). Historians looked for a definition for decades, once one campaign was launched in 2009, requiring historians to describe their academic activities in a day, thus based on these descriptions, an ideal definition was expected to be determined. Even though all users, including scholars, students, and others, suggested definitions based on their own experiences, none of them encompassed the entire concept of digital humanities.

Therefore, scholars already believed that there was no appropriate definition for this large concept. Later, the term *big tent*¹ emerged, meaning that because digital platforms contain numerous diverse disciplines, the whole concept cannot be described with only one definition. As a result of discussions, the name “digital history” was swapped with “digital humanities”, and the first term was just used for the ones who were working in history-related professions. Being a part of digital humanities was fascinating because, in this way, scholars were given a place that demonstrated that they belonged to this academic system. Before this happened, scholars felt as if they were at the wrong address. In the 1980s, the use of computers had already become common in the field of the humanities. William McCarty, the professor of humanities, defined the tech-oriented progress as a stain hindering his career (Crymble 2021, 6); however, his colleagues already provided a condition for him, in which he could feel a sense of belonging. A debate about the definition was held in the blogosphere in 2010, in which the expression “hack versus yack” was used (Crymble 2021, 6). In this definition, “hack” describes those who work with software applications, and “yack” refers to those whose jobs were theoretical; however, both sides claimed that the word “digital” defines their professions. Unfortunately, people who were working in this complex field made a common mistake, which was an attempt to apply a lot of meanings to just one word, leading a chaos to happen among scholars, thus they came together under the name of “digital history” which made impossible to understand what exactly each scholar does or did in this field since their professions differ from one another to a certain extent. In the past, each history-related field had a special name that could describe the separate part of the field, such as historical interpreter, librarian, historical journalist, editor for historical publications, genealogist, curator, archivist, and so on. Unfortunately, it is not possible to see this case on digital platforms. During the discussions on this problem that has become a deadlock, it was realized that scholars did not know about each other’s profession, and that was the source of the issue. Whenever they tried to find a proper definition, chaotic misunderstandings emerged because of their ignorance about each other’s professions. For instance, British and Irish historians might expect the new historians to use computer in the same way as the experienced historians use, however if they use the technological opportunities according

¹The “big tent” phrase was first used as the theme of the 2011 conference of the Alliance of Digital Humanities Organisations (ADHO), held that year at Stanford University and brought together scholars from diverse disciplines. Adam Crymble, *Technology and the Historian. Transformations in the Digital Age*, (University of Illinois Press), 2021, 6.

to the new traditions like in the book “Digital History” (2005), confusions are bound to occur since the works that they produced were very different from each other (Crymble 2021, 7). These unpleasant misconceptions affected scholars’ careers and led them to recreate their agendas under a broader scope. The educational system in the U.S. universities caused organizations in the education field such as the American Historical Association (2015) to change their guidelines and make the process more transparent, which showed that practitioners should pay attention to the way they communicate those differences and differentiate the activities that were supposed to be made on digital platforms (Crymble 2021, 8). In the book “Technology and the Historian”, it is explained how history and technology are connected, causing historians working in different ways to come together (Crymble 2021, 8). It is possible to find narratives written since the early 2000s regarding the scholars’ thoughts and experiences who were from North America and Europe, such as Chad Gaffield, Lara Putnam, Edward Ayers, Jane Winters, Tim Hitchcock, William G. Thomas III, and so on (Crymble 2021, 8). They have described how technological tools affected their career, research, and the exact field of history they worked in. The researchers of digital history attempted to show that there is a connection between digital history and traditional history, which means that digital history is not an independent notion. The goal was to make digital history a part of traditional history in a way that historians could understand this new advancement. Making this integration, history would be brought on the same level as other humanities-related fields, which were incorporated with technology. The book “*Technology and the Historians*” also contains the writings of well-respected thinkers such as Espen Aarseth and Janet Murray, reflecting the interaction of technology and history. It should also be remembered that this new field was not only under the influence of philosophy and theory but also each new tool of technological advancements that came from Silicon Valley. The ignorance of this fact would cause the hindrance of opportunities that could emphasize the influence of the reflection of technology on the field of history and the attraction of those who are interested in history. This book is considered a reminder to digital scholars. Being aware of the history of your field will help you to know what work has been performed before and how the future can be influenced by all previously made research. Without analyzing the past works, the future of history on digital platforms is unpredictable. By doing so, the works will not be forgotten, and thus the experiences and thoughts will be shared using a common language that everyone can understand. There exist several types of relations between history and technology, so it cannot be analyzed only from one aspect, but from

many. This means that historians should make a schedule that they can find separate time for each work they should implement in different places, such as the lab, library, lecture hall, special time for their own study, and so on. All of these phases of history are connected to the developing technology in different ways. Researchers' approach to history in a digital context is different; therefore, it is not right to look through the digital history in chronological order to make a narrative regarding the connection between history and technology, which could mislead readers. This is the reason why colleagues in this interconnected field are unable to find a proper definition, and the book, instead, just focuses on five aspects of history-related works provided by historians, which are historical research, the archive, the lecture hall, a learner-driven environment, and academic communication systems. The purpose of the book was to bring together all historians and digital scholars in a way that they can communicate and understand each other based on a common narrative and vocabulary related to both sides. This is why the book is considered a guidebook that they all can benefit from to analyze the effects of the digital age on this field. This book is suitable for everyone who has taken a step in this field, from students to senior historians, allowing them to explore the innovations in the field by exploring key changes and topics that describe what historians work on. Therefore, this is an ideal choice of book to lecture in classes, helping students understand the phases of historical works such as writing academic works, the use of digital archives, and digital tools for teaching. Another benefit of this book is that it is possible to make comparisons between the understandings of digital effects on history in different countries, meaning that the way people think about digital history may differ based on territorial standpoints. For instance, the approaches to the digital sphere of history might be different in Los Angeles and London, meaning that learners should search different sources from Roopika Risam to Domenico Fiormonte (Crymble 2021, 11) to explore more about the power of digitizing history, although the book "Technology and the Historians" consists of the works from the limited countries.

The first Chapter of the book is called "*The Origin Myths of Computing in Historical Research*" (Crymble 2021, 11) in which two methods have been described on how digital technologies have been used by scholars to work on their research during the 20th century and how historical works are affected by those tech-related opportunities in the 21st century, especially by Cliometrics which contains the topics of quantitative history and Humanities Computing. Historians applied some ideas from these two related approaches to handle the innovations observed in the 1990s, such as the increase in the use of the

Internet and massive digitization of historical works, which led to new challenges and opportunities to be pondered in more creative ways. Even though these two approaches were produced in the same period and based on computer-based works, they influenced diverse cognitive shifts and provided answers for different questions. One point that should be remembered is that even though all historians use technological tools, this does not mean that they serve the same purpose.

The name of Chapter 2 of the book is *The Archival Revisionism of Mass Digitization* (Crymble 2021, 12), which investigates the reasons why archives were digitized, analyzing different approaches from both the U.S. and the U.K. The reason for digitizing archives was simple – facilitating the use of sources in the lecture halls – whereas in the U.K., scholars were aiming to upgrade the level of research, which shows that when it comes to digitizing the historical records, the territory and the needs of scholars there play important roles. The technological tools also encouraged the scholars to shift their methods of working, inspired by the Web, virtual reality, crowdsourcing, conversion of data into sound, and so on. Thus, the new phase of sharing historical works started to be used, which was to put an end to the fixed structure of historical sources and make it more changeable and interactive.

In Chapter 3 – *Digitizing the History Classroom* (Crymble 2021, 12), the observations were made with regard to how technology diffused the education system and started to be used in classrooms. Four main topics were analyzed: quant-based strategies, providing collection, analyzing large data, and reconnecting with historical records. At least 130 study plans were looked through, and it was shown that the practices implemented in history were derailed by technological advancements, affecting the quality of the lectures and essays that were assigned to students. According to the research inspired by Silicon Valley and theories regarding education and psychology, the majority of the advancements in classes were not related to the advancements in using technological tools. The key point is that the way technology is used in classrooms differs due to regional preferences. For instance, in the U.S. and Canada the main goal was to teach students the methods to create digital narratives while in the U.K. digital tools were used to develop the tech skills, which allows us to see different approaches in education systems of countries and shows that it is crucial to know about the perspectives of diverse nations when it comes to using technological tools in education.

Chapter 4, titled "*Building the Invisible College*" (Crymble 2021, 12), examines how educators acquire tech skills through informal methods that were discovered to impart new

abilities that universities were expected to teach them in the distant future. This explains the reason for calling this chapter “invisible college”. Scholars participated in various activities, including summer schools, workshops, debate groups, and instructional videos. Some digital platforms were used by those who wanted to develop more skills that universities did not teach. Experts from other fields, such as linguistics, computer science, archives, and library science, had a great impact on the learning process of digital historians. However, there were still some issues that must be addressed. The main problem is that the innovations in the tech field were increasing, so there were always new advancements to study and work on. Therefore, the workload was too much, and newbies felt overburdened.

Chapter 5, “*The Rise and Fall of the Scholarly Blog*” (Crymble 2021, 13) shares information about the way historians communicated with one another and used blogging for that purpose. Blogging was one of the main digital achievements that drew people’s attention from the history-related occupations in a way that even digital communication platforms brought the most experienced and novice historians together. Blogs were the communication way preferred the most among historians, and as time went by, digital journals became outdated. At first, experts hesitated to use blogs because they looked more traditional rather than the transition from conventional to advanced. Using blogging became a common habit among historians starting in 2005. The first topics that were discussed were the academic problems encountered in universities. They did not exchange information about study outcomes. The use of blogs led experts in the field of history to be called “digital humanists”. Blogging had more social impacts on people than was expected to be more technological.

The name of the final Chapter is “*The Digital Past and the Digital Future*” (Crymble 2021, 14), which mainly explains that historians can form the future better by analyzing the role of digital tools in their professional lives, for example, the benefits of digitizing education, etc. In addition, historians should use a broader vocabulary to explain digital activities they perform in their field. The main idea was that because the preferences related to the use of technological tools based on the countries were different, the scholars had diverse experiences following the expectations in their region, which shows that specific solutions must be searched for the specific problems worldwide. Thus, the development of technology in education systems is possible, which means that it is a global matter and not specific to just one country or a group of countries.

2. What is NLP

Looking through Hammad's article regarding the history of NLP, we can acquire comprehensive information about the development of natural language processing. NLP stands for Natural Language Processing, which is a mix of linguistics and computer science. Being a field of Artificial Intelligence (AI), NLP serves to build communications between technological products and humans in accordance with human qualities. It is observed that the first conversations about NLP began in 1916, with Ferdinand de Saussure's coworkers, who was a Swiss professor, publishing his book (Hammad 2023, 2). He stated that what is expressed in a sentence depends on the relations among the words and their roles together with the whole context, which also means that the meanings of words are formed within a certain context, with the effects of qualifications that help them differ from other words. This was the fundamental knowledge about today's NLP. Chomsky took a further essential step in 1957, which was the introduction of the grammar book called "*Syntactic Structures*" (Hammad 2023, 2). With the help of this book, the analysts could figure out more clearly how machine translation could be performed. Chomsky aimed to create a few rules that could be used to determine the grammatical structure of sentences based on a specific language. Semantic nets which are techniques (nodes and edges) serving to describe semantic meanings formed in sentences, instead of translating each word separately used by Richard H. Richens in 1956 for the first time (Hammad 2023, 2). In this method of translation, nodes are themes, and edges are connections between themes, and together they are called semantic networks. Victor Yngve ran a set of algorithms to employ the grammar structure for phrases to generate meaningful sentences in 1960 (Hammad 2023, 2). Robert F. Simmons and Sheldon Klein, who were inspired by Victor Yngve, used the technique FOL, which stands for first-order logic, aiming at applying semantic nets (Hammad 2023, 2). There are four main elements in this method: the first one is "Predict," consisting of descriptive words used for expressing the object. The second one is "Term," by which objects are called or referred to. The next one is "Quantifiers" that allow us to address objects in whole or part. The last

element is Logical Connectives”: AND, OR, NOT, IMPLIES, and IF AND ONLY IF that serve to connect statements. The grammar structure based on the cases was used by Charles J. Fillmore, an American linguist, in 1968, which explains that each word in a sentence is linked to a specific case (Hammad 2023, 2). The cases are also called semantic roles, which can be Agent, Object, Beneficiary, Instrument, and so on. For instance, in the sentence “Jane gave Mike the book,” Jane is an agent, Mike is a beneficiary, and the book is the object (Hammad 2023, 2).

The first chatbot was programmed by Joseph Weizenbaum at MIT in 1964, which was called ELIZA, which used the simulation technique to build a communication between humans and machines (Hammad 2023, 2-3). The most applied script was DOCTOR, which played the role of a Rogerian psychotherapist and repeated users’ inputs as questions to show that it listens to them and comprehends what they say (Hammad 2023, 2). ELIZA was created using the programming language called MAD-SLIP, intended for conversational uses (Hammad 2023, 2). Although ELIZA failed to meet expectations, it was a significant improvement, which led people to boost their hope for a more technologically advanced future. A special organization called the Automated Language Processing Advisory Committee (ALPAC) was established by the National Research Council (NRC) in the U.S. in 1964, which intended to develop research in the field of NLP and AI (Hammad 2023, 3). Unfortunately, a setback in this tech field was inevitable since machine-based translation was far more expensive compared to human translation, and there was not enough storage space and resources. The speed of computers was not satisfactory either. The theory of conceptual dependency was introduced in the field of AI by Roger Schank at Stanford University in 1969 (Hammad 2023, 3). The model was programmed in a way that can describe the input information in computers. Particular attention was paid to the point that the meaning of the sentence should not depend on the words in the input, so that two sentences which have the same meaning could be represented in the same way. Another goal was to produce logical outputs. New steps were taken regarding the development of NLP around the world, regardless of the low quality of resources. Initial efforts were made under manually written instructions; however, NLP systems of that period were not capable of handling these complex tasks. In the 1980s, machine learning algorithms that were introduced in NLP launched a new phase in the improvement of AI (Hammad 2023, 3). Thanks to new methods, NLP models have become able to learn from data, which has led NLP systems to be more efficient and skilled. As NLP systems improved, larger datasets started to be trained, thus the theory introduced by

Chomsky (Hammad 2023, 3) became less popular gradually since it was based on language rules and did not support the idea of the use of corpus linguistics, which is the vast collection of texts. Corpus linguistics is crucial for NLP models to learn from data (Hammad 2023, 3). A series of machine learning translation tasks are highly used in NLP, such as sentiment analysis, text classification, named entity recognition, spam detection, information extraction, and so on (Hammad 2023, 3).

Recurrent neural networks (RNNs) and Long-Short Term memory (LSTM) are the networks used the most in NLP tasks (Hammad 2023, 3). RNNs are suitable for short-term dependencies and are based on the transformation of data through the layers from input to output layers. While in processing, this network structure can remember the previous information from the short-term memory, which was transformed via the past layers. There are two types of RNNs: finite impulse response (FIR) and infinite impulse response (IIR). The main difference between FIR and IIR is that FIR is based on a limited number of inputs, and the old data is forgotten after a while, while IIR always keeps providing inputs in the short memory and is based on an unlimited number of prior inputs. RNN is considered for tasks such as speech recognition and machine translation, speech-to-text, text-to-speech, predicting stock prices, weather forecasting, text generation, and so on. LSTM is the advanced version of RNNs, which focuses on the problem of vanishing gradients, which serves to update the weights of layers every time new information is learned. What makes LSTM different from RNNs is that LSTM is a type of network that is able to memorize information in the long-term memory. However, LSTM has a slow and complex structure; because of that, Gated Recurrent Units (GRUs), which are the simplified version of LSTM, are also used. GRUs have two gates that shape the work of this neural network: the Update gate and the Reset gate. The Update gate chooses necessary data to keep in mind, while the Reset gate is responsible for identifying unimportant data to be deleted. Both LSTM and GRUs are available for a wide range of tasks in different fields such as sentiment analysis, question answering, named entity recognition, traffic flow prediction, energy consumption modelling, sign language translation, DNA/RNA sequence modelling, and so on (Hammad 2023, 3).

In the 1980s, Geoffrey Hinton programmed a neural network called Boltzmann, used for language modelling (Hammad 2023, 3). It is a type of statistical model that was employed for the prediction of the probability of the word order, which is targeted to learn the statistical relations between words. It was first trained on a collection of text documents. As an example of this technique, n-grams language models were used. This

method was based on the idea that the occurrence of a word depends on what the previous word is. Unfortunately, statistical models were not satisfactory, since their qualifications were not enough in terms of working on long-range dependencies (Hammad 2023, 4).

Yoshua Bengio suggested a new method of language modelling in the 1990s, which is called Feedforward Neural Networks (FNNs), capable of learning sophisticated relations between data (Hammad 2023, 4). In the structure of FNNs, there is no memory to keep data since loops are not typical for this neural network. Data flows in the forward direction, and this language model is suitable for static data, which is not sequential, unlike RNNs, which are used for sequential data and loops. Transforms, on the other hand, have become surprisingly handy because of their advantages. Transformers are another type of neural network, and likewise, they do not process data sequentially; instead, they use a self-attention mechanism, which serves to identify necessary words in sentences to be processed. This mechanism provides a score that shows how much attention should be given to each word. Here, an important nuance is that Transformers use the system of multi-head attention, which is a collection of a series of attention mechanisms. Each head is applied to learn from different parts of the input. Because transformers do not use sequential data, positional encodings are added with the purpose of receiving information about the positions of words in sentences, which allows the language model to distinguish between the positions of different words. One of the steps in the application of transforms is to use FNN layers to process data in more detail. Furthermore, Transformers have two important parts, which are called the encoder and decoder. In order to encode the input sequence, an encoder that encompasses several layers of attention mechanisms and feedforward networks is used. Each encoder layer processes the input data by applying self-attention and feedforward networks, while the decoder part is applied to generate the text, which uses one more attention layer for the output received from the encoder part. The key difference of the decoder is that it prints predictions using the encoder's output. Transformers and feed-forward neural networks play vital roles in forming large language models.

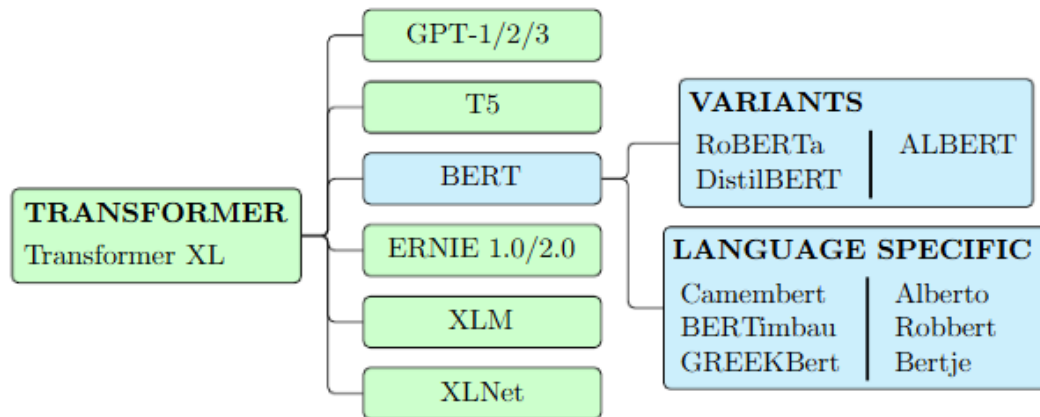


Fig.1. The evolution of Transformer-based models²

In this graph, Transformer XL is an extension of Transformer, specifically designed to handle long-range dependencies, while the others are models built using Transformer (Hammad 2023, 4).

Word embeddings are a way of representing texts in which they are described through vectors in an n-dimensional space (Hammad 2023, 4). Tomas Mikolov coded a new type of neural network in the 2000s, which is called Word2Vec, that uses word embedding structures, which make it possible to discover syntactic and semantic relations among words, such as CBOW (Continuous Bag of Words) and Skip-gram, which are two architectures for the representation of word embeddings. The difference between these structures is that CBOW predicts the word from surrounding words and context, while Skip-gram, on the other hand, predicts the context from the given word. Like Word2Vec, GloVe is another type of representation of word embeddings based on methods such as the use of global co-occurrence of words, the creation of a word matrix, and the factorization of the matrix. As can be seen, it follows a different architecture from that of Word2Vec (Hammad 2023, 4).

All these advancements in NLP led scholars to store a large amount of data to use for training later, applying different language models. Thanks to these developments, NLP models are capable of performing complex tasks such as question answering, text summarization, and machine translation more precisely, which in general were crucial steps towards the production of AI tools and products that can communicate with humans.

²Nikolaos-Ioannis Galanis, Panagiotis Vafiadis, Kostas-Gkouram Mirzaev, and George Papakostas, "Machine Learning Meets Natural Language Processing – The Story So Far," ResearchGate, 2021, <https://doi.org/10.48550/arXiv.2104.10213>

Google Translate, one of the most prominent applications, started to be used in April 2006. It is an unpaid and web-based service that is used around the world to translate texts in diverse languages. In order to serve this purpose, Google Translate uses the technique of statistical machine translation (SMT) based on the algorithms used for predictions. Even though the quality of this method is not high enough to provide accurate translations, it is still widely used (Hammad 2023, 4).

Another worth-mentioning progress is the presentation of GPT created by OpenAI, standing for Generative Pre-trained Transformer, which is a large language model that was trained on a large number of datasets to generate texts. Its upgraded versions were introduced later. For instance, T-NLG, created by Microsoft, standing for Turing Natural Language Generation, and GPT-2 performed even better, displaying great achievements such as being able to learn long-range dependencies, which helped to improve text summarization and text generation. What is more is that these large language models were suitable for use in chatbots to improve the quality of communication between humans and AI tools (Hammad 2023, 4).

According to the results provided by Google Scholar and Scopus, papers related to NLP published since 2011 rose consistently.

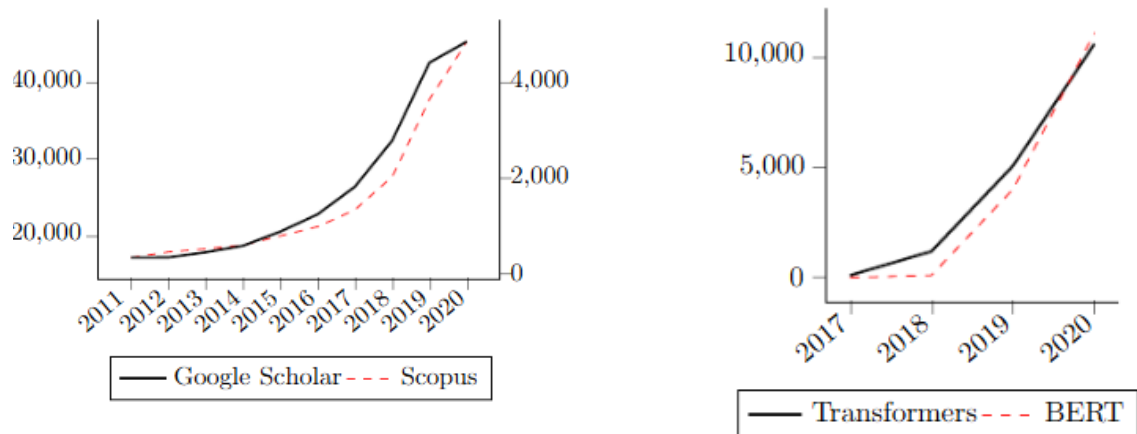


Fig.2. Publications for NLP in general and references to popular Transformer papers³

³Galanis et al., “Machine Learning Meets NLP,” 3.

As can be seen from the graph above, the index of publications is increasing throughout the years on both platforms. Especially, after the use of Transformer models such as BERT, the index of the publications boosted significantly (Galanis et al. 2021, 3).

There are several NLP tools and libraries in use, which are free to apply and differ from each other based on their qualifications:

	Stars	Description
spaCy [44]	19883	Python library with pretrained models and great multilingual support, not suggested for research and benchmarking.
Flair [45]	10091	PyTorch library developed by Zalando, featuring the “Flair embeddings” for more efficient text vectorization.
AllenNLP [46]	9794	Designed for quick prototyping and research with a variety of pretrained models.
NLTK [47]	9711	One of the most recognizable libraries for NLP with some ML options - though not a common option for the task.
Stanza [48]	5271	Stanford’s Python library for “Many Human Languages”. Can also be used as an interface to CoreNLP for even more features.
SparkNLP [49]	2002	Built on top of Apache Spark and TensorFlow for speed and scalability, with generic and domain specific models available.

Table 1: An overview of popular open-source NLP tools & libraries⁴

For instance, NLTK is very useful to teach the process of analyzing data step-by-step, while spaCy and Stanza are used in real-world projects. The difference between spaCy and Stanza is that spaCy is faster and has low-level complexity, while Stanza is a medium-level complex library, but available for more than 70 languages, which is an advantage compared to spaCy because of its limited language opportunities. AllenNLP, on the other hand, is used to conduct research and recommended for tasks such as text classification and Question Answering. This library also has limitations for languages and is not useful for simple tasks, since it is designed for more complex tasks. Flair is a great library used for conducting research and also employs the technique of transfer learning with text embeddings. This library is good at performing several tasks such as named entity recognition, part-of-speech tagging, classification, and so on. As for SparkNLP, this is considered for use in workplaces for tasks: tokenization, sentiment analysis, text classification, dependency parsing, and named entity recognition.

⁴Galanis et al., “Machine Learning Meets NLP,” 7.

Even though NLP is gradually increasing, there exists a variety of problems to handle. As an example, unstructured data makes data processing difficult since the majority of the words are not necessary. For that, the data should be cleaned of ineffective elements, and then the main algorithms can be used to explore the data and achieve the targeted output. Data consisting of texts should be split into sentences, words, and then their base forms to make it useful for the research. Unfortunately, together with noise words, there are other challenges that NLP tools may encounter very often. Ambiguity is one of them, which means a confusion in figuring out the exact meaning of the sentence since it may have alternative meanings. Several types of ambiguity are observed (Galanis et al. 2021, 7-8):

- Lexical ambiguity is a challenge in which a word can be used in different kinds of part-of-speech tagging.
- Semantic ambiguity can be observed when multiple interpretations of a sentence are possible
- Syntactic ambiguity occurs when several interpretations of a sentence can be performed based on the availability of different kinds of parsing.

Irony, sarcasm, slang words, and inconsistencies are the common obstacles to be dealt with. Pragmatics is the field that analyzes the main concept of sentences regardless of the literal translations of words. It focuses on how people use the language to communicate with each other. It is possible to see challenges in NLP and different types of ambiguities that were encountered in the image below (Galanis et al. 2021, 8):

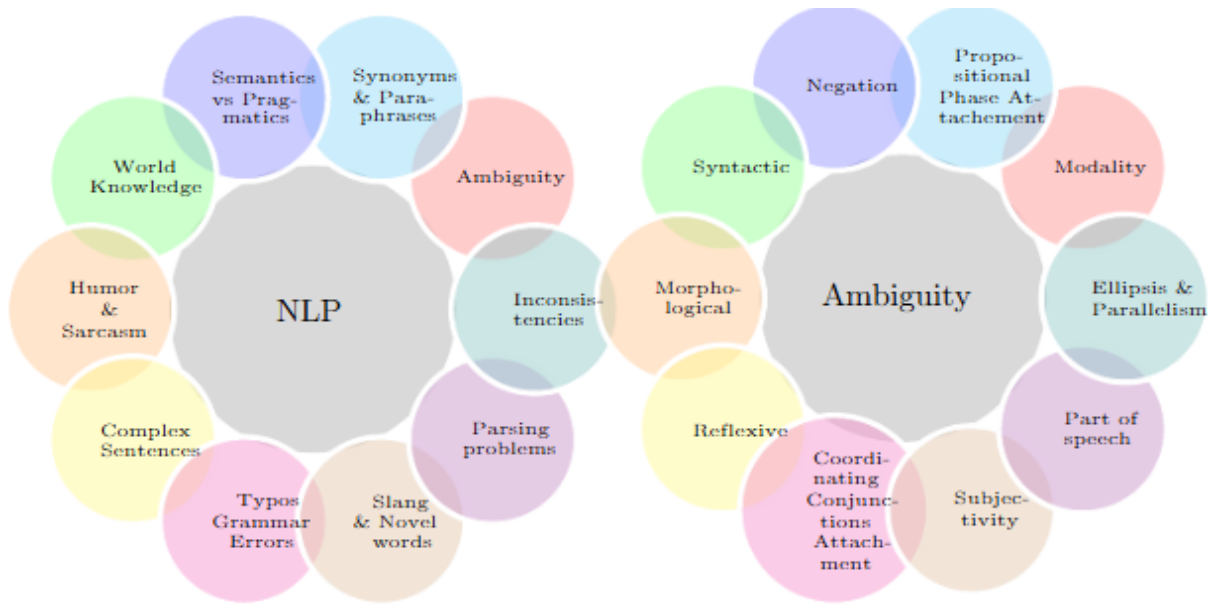


Fig.3. NLP Linguistic Challenges while parsing human languages⁵

Programmers work extremely hard to program new models consisting of billions of parameters whenever they introduce a state-of-the-art one to the public, since it possibly provides more precise outputs. Unfortunately, it is time-consuming and overpriced. Here is shown the number of parameters for each model in millions. Here is shown the number of parameters for each model in millions:

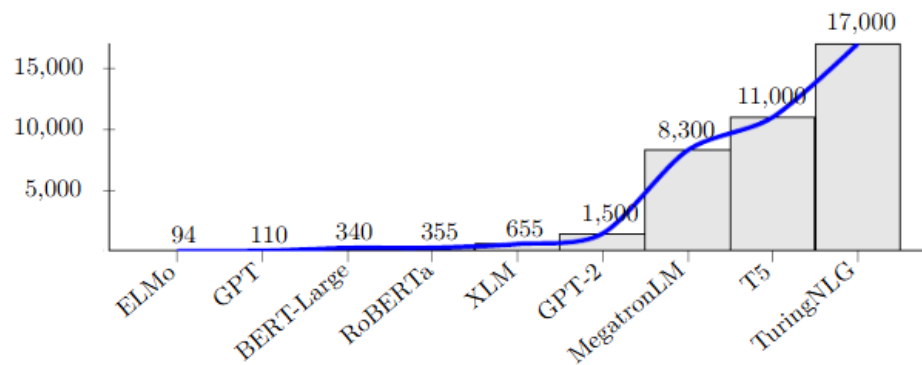


Fig.4. Number of parameters for models over time (in millions)⁶

⁵Galanis et al., “Machine Learning Meets NLP,” 8.

⁶Galanis et al., “Machine Learning Meets NLP,” 9.

During the last few years, the field of NLP has achieved considerable successes, from vectorizing data to the introduction of Transformers and transfer learning, which is known as the Golden Age. Machine learning methods and models are being implemented and improved day by day. Even with all these developments, scientists and analysts need to work harder to overcome the current issues, such as language modes being available for limited languages and a lack of sufficient hardware, and so on. Today's most-used language models are the ones provided by Transformers. Even though AI is capable of performing high-level coding, this field is still not able to meet expectations at its full potential and capacity. What is important is that there needs to be an approach to AI capabilities in a realistic way, and proceed with the improvement of the process in the AI revolution that the world is going through now (Galanis et al. 2021, 10).

3. NLP usage in/for historical research

The digitization of historical issues is beneficial for several reasons. To begin with, even though physical documents are damaged or lost, the digital versions will be available to be used and the second important nuance is that scientists from different parts of the world can have easy access to those documents for their task, so they will not need to travel all the way to receive them. Over the last two decades, a considerable volume of historical documents, including manuscripts, legal documents, literary works, and articles, have been scanned by organizations such as museums, institutions, and libraries. Then, all these records began to be transcribed into a usable format for digital tasks, such as data mining, information retrieval, and question answering, among others. Historical Document Processing uses lines of coding, algorithms and different software tools to process data. With this purpose, subfields of computer science, including NLP, computer vision, OCR, etc., have been used, which shows that it is a multi-source field. A few steps of the analysis should be performed during the process of historical records, which are the same for both printed and handwritten documents (Philips and Tabrizi 2020, 4).

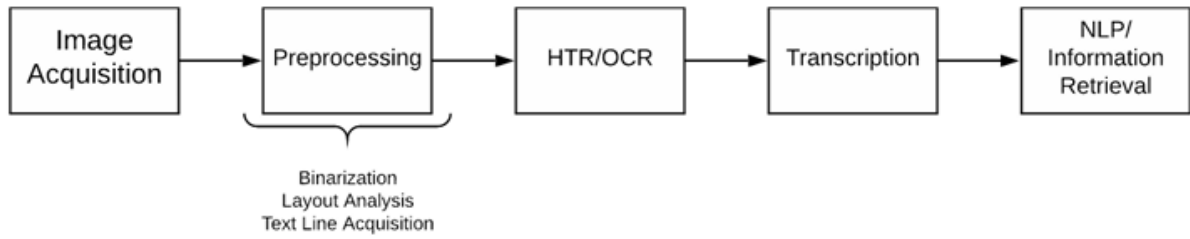


Fig.5. The steps in a conventional Historical Document Processing workflow for both handwritten and printed documents.⁷

As can be seen from the pipeline of the entire process based on the analysis of Philips and Tabrizi, the first step is to scan or have the document imaged, then to detect the text easily, binarization/Grayscale Thresholding has been applied in which the image is turned to gray and a threshold/numeric value is determined to clarify whether the lightness of image pixel is above or below the threshold. Based on the result, the image color will be white or black, respectively. Layout Analysis is the step that divides the document into sections such as headings, images, columns, graphs, etc. Text Line Acquisition serves to remove any distortions regarding the lines of text so that the document can be ready for obtaining more accurate OCR results. HTR and OCR stand for handwritten text recognition and optical character recognition, respectively. Neural networks are used in this step to perform character recognition. The purpose of this step is to convert the document into machine machine-readable format, such as docx, txt, etc. The fourth step is Transcription in which audio files are converted into text formats and during the process of this phase background noise is removed, the audio is divided part by part to be handled more efficiently and then some language models are used such as machine learning models that transcribes the speech into text and contextual language models for enhancing the accuracy of text's language. In the sixth step, NLP is used, where numerous NLP tools and algorithms are applied. The text went through the preprocessing stage at first, including tokenization, stopword removal, lemmatization, and lowercasing, then the text is converted into numerical values using word embedding representations, and linguistic analysis such as NER, POS, Sentiment Analysis, and Dependency Parsing are implemented. In the next step, NLP models are used and predictions are made. This round includes text classification, text generation, and machine translation. In the last process of NLP, long texts are summarized, and possible errors are corrected. In the final process of NLP,

⁷ James P. Philips and Nasseh Tabrizi, "Historical Document Processing: A Survey of Techniques, Tools, and Trends," arXiv, February 2020, <https://arxiv.org/pdf/2002.06300v1>, 3

Information Retrieval (IR) is performed. During this step, several types of algorithms, such as the creation of an index, are used to detect which words are used in which documents. Later, each document is ranked according to the user's query, which is matched with the index to retrieve information, and the order of results is adjusted using upgraded methods such as semantic research (Philips and Tabrizi 2020, 4).

Institutions that possess handwritten historical documents applied the technique of microfilming to maintain the documents from previous ages, which worked in a way that firstly photographed those documents and then reduced their size to store them for future use. Later, the steps leading to the digitization of documents were taken. Early English Books Online started to be used, which is a digital archive that contains the digitized versions of historical records such as pamphlets, books, and almanacs, etc., related to the 15th to 18th centuries. The main target was to provide tools that work accurately for layout analysis and transcription. The technological progress in printing documents was presented to the public by Johannes Gutenberg. Before his printing machine, the documents were copied by hand. With the invention of this machine, literary works started to be copied for publication; however, other documental jobs such as keeping records, the production of private documents continued to be prepared by hand. Structural differences affected the complexity of documents. For instance, the format of medieval manuscripts was more difficult than that of letters in the 18th century; however, letters were written in cursive style, and this made the transcription process of letters even more complicated. A lot of research has been performed on providing technological tools and models targeted for medieval documents, while the progress for ancient and early modern documents in the tech field has not been observed to the same level as medieval documents. The pictures of letters on both left and right sides and the speech draft at the center, which belong to the 18th and 19th centuries, show the complexity of the cursive writing (Philips and Tabrizi 2020, 4):



Fig.6. 18th and early 19th century letters (left and right) and speech draft (center) from the United States illustrating handwriting differences and challenges in archival documents.⁸

Below are images of the Papyri of Greek Biblical texts maintained in the Chester Beatty Collection on the left, and two medieval manuscripts in Latin at the center and on the right:

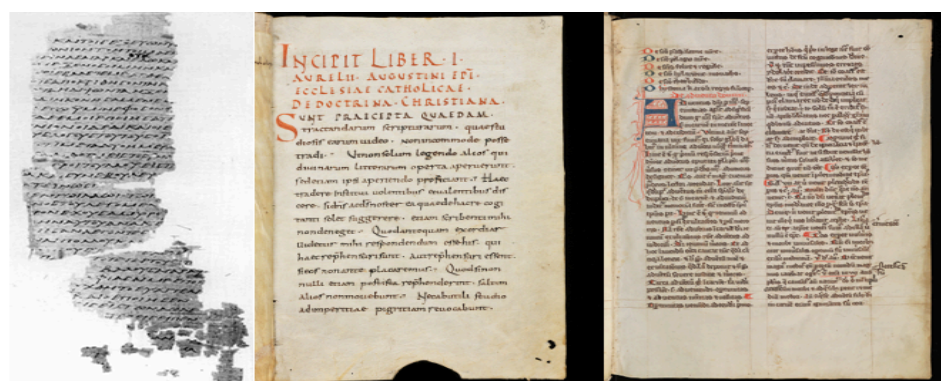


Fig.7. Fragmentary Greek Biblical Papyri from the Chester Beatty Collection (left²) and Medieval Latin manuscripts from the e-codices project (center³ and right⁴) illustrating complex document layouts.⁹

New techniques are being developed for the recognition of handwritten historical records by scientists using state-of-the-art approaches of neural networks. During the practices of new methods, two datasets (Washington and Parzival) were used, and the new approaches performed far better than Hidden Markov Models (HMMs), which were operated for the HisDoc project focused on character-level recognition. The accuracy of

⁸ Philips and Tabrizi, “Historical Document Processing,” 6.

⁹ Philips and Tabrizi, “Historical Document Processing,” 5.

the Washington dataset's output was higher by using neural networks in comparison to applying HMMs, which is basically a statistical model based on the use of hidden states and observations, namely visible and measurable elements in datasets. Granell and his coworkers operated Convolutional Neural Networks (CNN) to research medieval records since CNN is a type of neural network suitable for image classification. The disadvantage of these advanced models is that they need to be applied on high-standard computers because of the complexity of the new methods. In general, neural networks have become the most used techniques to process historical documents. As for Optical Character Recognition (OCR), neural networks are also recommended to be used. The main challenge for OCR is that typography in earlier ages was versatile and far more different than today's. The historical records from the 15th to 19th century were specific to those ages and they are now unidentifiable for today's OCR tools, which were trained based on currently available typography styles. Abbreviations, word-hyphenation, and many other specific features make old books and records impossible to investigate. With the intention of solving this issue, the researchers are advised to apply open source options such as OCRopus and Tesseract, which can provide outputs with high accuracy. Tesseract can be used for both simulated data and the analysis of character images, but both alternatives require a significant amount of data to provide manually created and verified annotations. For processing historical documents, some datasets are present following the tools and periods below (Philips and Tabrizi 202, 9-13):

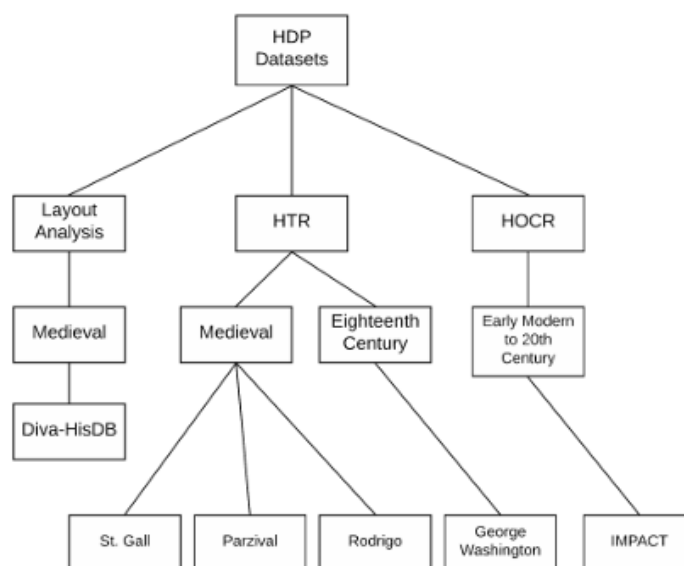


Fig 8. A Hierarchy of HDP Datasets Based on use case and time-period¹⁰

¹⁰ Philips and Tabrizi, "Historical Document Processing," 14.

As can be seen, for the medieval age, the datasets including Diva-HisDB, St. Gall, Parzival, and Rodrigo are available, while the dataset called George Washington is for the 18th century, and IMPACT for the early Modern to the 20th century. There exists a certain way of evaluating metrics suitable for each method to assess the performance. For instance, precision and recall are used for HTR. Precision measures the ratio of outputs that are precisely identified and relevant according to the consideration of the system, and Recall determines the ratio of outputs that are identified correctly. F1-score, which is the harmonic mean of Precision and Recall, should also be applied after the use of precision and recall. These metrics are specially available for tasks such as NER, Table detection, classification, Layout analysis, OCR, and HTR. For the evaluation of recognition tasks, Character Error Rate (CER) and Word Error Rate (WER) are applied too. As for the assessment of layout analysis performance, Line Error Rate (LER) and Segmentation Error Rate (SER) are also used (Philips and Tabrizi 2020, 14).

Another valid method to conduct research on historical documents is to perform Sentiment Analysis, which serves to identify if the message or reaction delivered in the text is positive, negative, or neutral. Unfortunately, the challenges of ancient documents can be observed while performing this task as well. For instance, the language of the texts may have changed to some extent, and the grammar may be different from modern one. Words may have different meanings or even not be used anymore. In addition, contexts may not be understandable due to cultural and periodic changes. If distortions or errors exist in documents, the analysis may be even more complicated. With the performance of sentiment analysis, it is possible to understand what happened in different fields of life in the past, including political speeches, rules, events, beliefs, rituals, revolutions, narratives, and so on. In general, sentiment analysis helps researchers explore the previous ages more accurately and profoundly.

4. A Case Study: gender norms on feminist writings (USA, 19th-20th centuries)

The purpose of this research is to investigate the reflection of gender norms on feminist writings between the 19th and 20th centuries. A few topics have been taken into

consideration per the periods: while *Seneca Falls Convention (1848)*, *Elizabeth Cady Stanton's speech and Trial of Susan B. Anthony (1873)* reflects the first wave of feminism, *Betty Friedan's The Feminine Mystique* and *Miss America Protest (1968)* are considered as the second wave. Possible answers will be searched for two questions: *How can natural language processing (NLP) models play a role in explaining the changes in the language based on gendered features, and how are those shifts represented in feminist works over the process of 2 centuries?*

By exploring this matter, it will be possible to see the status of women in society in the previous centuries. Besides, the connection between the first and second waves will be made, which has historical value since this will allow us to see the way these activities have been reflected in public discussions over centuries. In addition, NLP models provide new methods that can help to examine and discover what is dark for us.

Chapter 1. The First and Second Waves of Feminism

1.1. First wave of feminism

The first wave of feminism in writings is observed at the end of the early 19th century and the beginning of the 20th century, which was mainly focused on the legal, political, and educational issues. The first wave was promoting gender equality, justice, and women's rights. The main issue for this period was the right to vote and to be recognized as an independent individual. To support and fight for women's rights, the activists delivered formal speeches, published essays and pamphlets, filed petitions, and lodged appeals against unfair treatments of women. Some of prominent members of this first wave: **Elizabeth Cady Stanton**, *the president of National Woman Suffrage Association from 1892 to 1900*, **Susan Brownell Anthony**, *American essayist and civil rights activist*, **Mary Wollstonecraft**, *considered as the founder of liberal feminism and best known for her book *A Vindication of the Rights of Woman**, **Sojourner Truth**, *American women's right activist*, **Lucretia Mott**, *abolitionist and social reformer*, **Emmeline Pankhurst**, *British activist and politician*. They all played a significant role in defending women's status and rights in society.¹¹

a) Seneca Falls Convention

Elizabeth Cady Stanton and Lucretia Mott held the Conference of Seneca Falls Convention in 1848 in New York, which is considered the first step of the movement during the first wave of feminism in history. Before the Seneca Falls Convention, a

¹¹Guy-Evans, Olivia. "First-wave of feminism." *SimplyPsychology*. February 13, 2024, <https://www.simplypsychology.org/summary-first-wave-of-feminism.html>.

meeting by the M'Clintock family was held to determine the agenda of the upcoming conference. The participants of the meeting – Elizabeth Cady Stanton, Mary Ann M'Clintock, and her daughters Elizabeth and Mary Ann Jr. – listed the issues regarding women's roles in society that bothered them. They described equality between women and men in five main aspects: education, career, religion, family, and ethical conduct. The draft of the Declaration of Sentiments was prepared during this meeting. It was adapted to the US Declaration of Independence, making essential changes to some parts, for example, the expression of “the history of the present King of Great Britain” was swapped with “the history of mankind” and Elizabeth Cady Stanton also added the part “and women” to the sentence “all men are created equal”. In the meeting, Elizabeth Stanton noted the right to vote on the list. Even though her husband helped her during the meeting, he still hesitated to endorse this idea and did not even participate in the convention. Harriet Cady Eaton, the sister of Elizabeth Stanton, attended the convention with her son, showing the family's support. Lucretia Mott, one of the main organizers of the conference, attended with her ill husband, James Mott, and sister, Martha Wright. 300 people participated in the conference, and 40 of them were male. Frederick Douglass, American politician and abolitionist, was one of the attendees. In this conference, vital subjects such as social, civil, and women's rights and religion were discussed. The Declaration of Sentiments, which starts with the statement regarding the equality between woman and man, and a list of resolutions were introduced, in which legal rights, the right to vote, and more were demanded, the list of 18 complaints about the lacking rights related to the different field such as education, career, and property rights was presented. The main point of the Declaration of Sentiments was to win the right to vote for women. Even though some participants hesitated to support this demand, Fredrick Douglass voted for the acceptance of this requisition. At the end, 100 participants, 68 women and 32 men, signed the document. Even though the progress was not fast, the right to vote was ratified in 1920, and the Seneca Falls Convention played a fundamental role in achieving this success. The meetings, organizations, and protests for the sake of equality increased. The Seneca Falls Convention managed to create awareness concerning the need to fight for women's rights and ensure their permanent progress in every aspect of life. Following this conference, the Rochester Women's Rights Convention

in Rochester and the first season of National Women's Rights Conventions in Worcester, Massachusetts, were held, which shows that such debates will be going on in the future.¹²

b) Elizabeth Cady Stanton "Address on Women's Rights"

In her speech "Address on Women's Rights" delivered in 1848, Elizabeth Stanton criticizes the problems related to gender inequality, including in every aspect of life, caused by misinterpretations of religious documents and social norms aiming to justify the obedience of women. Stanton states that the religious narrative, such as the lives of Adam and Eve, has been taken advantage of to serve their intended purpose – to ensure the male dominance over society. In these misinterpretations, people believe that they are all true women are described as the personification of sins, while men are considered morally pure and innocent. Unfortunately, this situation caused women's self-esteem to be damaged. Besides, there is a stereotype that men are more powerful than women when it comes to muscular strength. In order to make a comparison about this matter, both genders should be trained in the same way from the very beginning. People have physical superiority only when they have physical power, willpower, and endurance together. For example, the horse is stronger than a man physically; however, it is the mind that man uses to control the horse, so people have more important qualities than physical strength, which are mind and willpower. To enlighten the population, Stanton interprets the religious stories again and highlights that women and men are equal in the eyes of God; thus, women should not be suppressed under the patriarchy. Oddly enough, while men are allowed to cause and commit violence against women, these victims are subject to living under these unfair conditions. Stanton also emphasized the fact that the leaders of religious organizations and institutions do not endorse the idea of women getting educated and participating in social or political reforms. Education is the key to gaining independence from the oppression of male dominance. Therefore, women are not allowed to receive a high quality of education so that they can finally realize that they should attend the meetings of legal and political platforms and vote for the laws that will shape their society. However, women are obliged to pay taxes for the institutions that only men can benefit from. The exclusion of women from the political sphere does not affect only the process of voting; this situation also

¹²Mindy Johnston, "Seneca Falls Convention." *Britannica*. Accessed on May 5, 2025. <https://www.britannica.com/event/Seneca-Falls-Convention>.

frames women as second-class citizens of society, which undervalues women's dignity and humanity. However, Stanton believes that the legal systems have much to benefit from the experiences, insights, and intelligence of women. She proves this fact through a few examples from history. For instance, even though there was political tension between the Arab queen Zenobia and her enemy Emperor Aurelian, he admitted that she is powerful and has a strong character. Margaret, the Queen of Denmark, united three kingdoms: Denmark, Norway, and Sweden. Isabella of Spain was a very well-known queen thanks to her intelligence and leadership skills. The history is full of wise and powerful queens, which proves that women should not be underestimated if the development of the country and its society is the subject of the discussion. Another vital issue is that when women get married, all the rights concerning their properties are taken over by their husbands, which shows that women become deprived of their property rights. As a response to this unfairness, Stanton demands that legal forms be held for women to obtain the rights they deserve, thus they can sign contracts, make their own decisions over their properties and lives. She also criticizes the legal system for not supporting women's higher education. Because they are taught just enough knowledge to be a wife and mother, they are not able to gain their independence from male dominance. However, it is a fact that they deserve to get an education at the same level as men. Elizabeth Stanton strongly supports the idea that women and men who support women's rights should come together and act in solidarity to win this struggle, erase the oppression of men, and ensure equality in each sphere of life, including social, political, educational, legal, and career.¹³

c) Trial of Susan B. Anthony

Susan B. Anthony was arrested in 1873 for voting in the 1872 elections illegally by opposing the state law that only allowed male citizens to vote. However, she claimed that she had the right to vote based on the Fourteenth Amendment of the U.S. Constitution, which exists a part that no state shall enforce any laws that constrain the rights and immunities of the state. Anthony voted together with her fourteen female neighbours, such as Ellen S. Baker, Nancy M. Chapman, Hannah M. Chatfield, etc., and before being arrested, she delivered encouraging speeches, one of which was very famous called "*Is It a*

¹³ Elizabeth Cady Stanton, "Address Delivered at the Woman's Rights Convention, Seneca Falls, New York, July 19, 1848," Susan B. Anthony Papers, September 1848, <https://susanb.org/wp-content/uploads/2018/12/Elizabeth-Cady-Stanton-Sept.-1848.pdf>.

Crime for a U.S. Citizen to Vote?” across the country. One of her speeches was published in a newspaper with a full version. The judge, Ward Hunt, who was infamous for opposing women's suffrage, directed the jurors to find Anthony guilty without discussing the case and fined Anthony \$100, but she refused to pay the fine. The judge stated that she will not be put into jail because of this rejection, making this decision, he prevented her from applying to the Supreme Court for her case. This action of the judge was then criticized, and a circuit court judge said that it was not right for a judge to misdirect a jury to declare a defendant guilty without investigating or discussing the case. On the last day of the judgment, the judge let her defend herself. She reminded them of a fact that, just like men, women also have natural, civil, political, and judicial rights. Unfortunately, society still treats men from white collar to drunk as if they are superior, which means that if women are not treated equally with men, they have no right to judge women. Anthony's friends were also arrested but put on trial. As a result of the trial, they were found guilty of violating the state law and fined; however, they refused to pay the fine and were put in jail. The inspectors who allowed women to vote were arrested but pardoned by President Ulysses S. Grant later. Before registering the votes, those inspectors consulted a well-respected lawyer of that time, John Van Voorhis, who was a strong supporter of women's rights. After voting, Anthony was interviewed by a newspaper, and once her interview was released, the call to arrest the inspectors was made. Susan B. Anthony's trial was a turning point in women's struggle for suffrage. She fought for justice, building a unity of supporters that continued even after her death. Rights defenders of today follow in her footsteps.¹⁴

The suffragist couple, Francis and Virginia Minor, from Missouri, came up with a new strategy, called New Departure, which was based on the 14th and 15th Amendments, aiming to recognize people who were deprived of their voting rights as citizens who are already allowed to vote. The Fourteenth Amendment defined the word “citizen” in a more general scope, which includes women. The districts used to possess full control over the voting process, but with the ratification of new amendments, a federal circuit court started to supervise the control of voting rights in part; thus, the federal court ensured the protection of the rights. The New Departure strategy intended new principles that are in favour of women's rights to be verified through the court. National Woman Suffrage Association (NWSA), founded by Susan B. Anthony and Elizabeth Cady Stanton,

¹⁴“Susan Anthony Trial (1873),” School of Law (UMKC), <https://famous-trials.com/anthony>.

officially adopted the New Departure strategy in early 1871, which led women to use their right to vote, and in case of any rejection, the official complaints would be made to the court through lawsuits. Soon, numerous women tried to use their right to vote, however, sixty-four of them were unsuccessful. Then another large group of women attempted to vote, but they were also rejected. A complaint was made to the Supreme Court in the District of Columbia. However, the Court issued a decision against women, saying that having a citizenship does not give the right to vote, and the intelligence level of the current population is not appropriate for participating in elections. The court also stated that even if they had used this right, that would have left the civil government in ruins. However, Anthony was able to vote in the 1872 election, possibly because she was a very well-known public figure. She even founded the first political organization of women together with Elizabeth Cady Stanton in 1863, which is called the Women's Loyal National League. Anthony managed to collect 400,000 signatures for the petition against slavery. Apart from that, she and her colleague Stanton ruled the National Woman Suffrage Association.¹⁵

After the legal trials were over, Susan B. Anthony wrote a book, describing every possible detail about the process, such as the provocation, her speech during the trial, what the attorneys talked about, the written documents of the trial, and how the judge acted during the trial. She was unable to add the arguments made by the U.S. Attorney Crowley because he did not allow it. There was also an essay in Anthony's book written by John Hooker after the trial, he was the reporter of the court in Connecticut's Supreme Court. As a reporter for stating the errors made in the court, he noted in his essay that Judge Hunt violated all the rules of controlling a trial under the state law.¹⁶

Susan B. Anthony was such an appreciated public figure that even in the entertainment industry, the entertainers personified her and delivered her brave and encouraging actions to audiences' attention.

1.2. Second wave of feminism

Second-wave feminism was observed starting from the 1960s and continued until the 1980s. Unlike the first wave, the second one constituted the meetings, movements, and

¹⁵ "Trial of Susan B. Anthony", *School of Law (UMKC)*

¹⁶ Ibid.

more significant goals were set. Second-wave feminism aimed to change the old-fashioned mindset by altering the cultural, political, and economic systems that may give rise to any kind of gender discrimination. During this period, supporters and activists of this wave fought to solve numerous social issues such as domestic violence, sexual harassment, and discrimination in the rights at workplace. One of the main issues of this phase was to give women the right to choose their bodies, which includes matters such as abortion and birth control. A landmark decision, which is called *Roe v. Wade*, was adopted, giving women the right to have an abortion. This victory was a considerable progress for women. Another challenge was to receive fair treatment in the workplace. The discrimination was observed from different aspects, for example, female and male employees did not get paid equally, and women could not get into the position of leadership at work. It is possible to analyze the second wave of feminism through the meetings and literary works.¹⁷

a) Betty Friedan's The Feminine Mystique (1963)

The *Feminine Mystique* is a book written by Betty Friedan, which is about changing the outdated view of women's roles in society. In the past, it was believed that women had only one role in society – being a housewife. Friedan was challenging to alter this unfair treatment against women by using her book. The book describes the lives of unhappy women between the late 1950s and early 1960s, introducing the issue as “the problem that has no name”. The *Feminine Mystique* consists of 14 chapters and starting from the first one, the social issues of that period are clearly expressed such as increase in childbirth, women's dropping the college, the decrease in the average age of marriage and the conclusion of this chapter shares a significant message meaning that women do not just require of having the life of housewife but they also want to be a free member of society and participate in public platforms. In the next chapter, Friedan noted that women's magazines were prepared by men who were misleading people by insisting on the stories and articles that were intended to show that women are happy as housewives but unhappy as free individuals, building their careers. In this way, they created the idea of the feminine mystique, which explained that women prove themselves by fulfilling their duties as housewives and devoting their entire lives to their family. Friedan also highlights the fact that this case was not even a matter of discussion in the 1930s because women's magazines

¹⁷“What was the Second Wave Feminist Movement,” *DailyHistory.org*, July 20, 2021. https://www.dailyhistory.org/What_was_the_Second_Wave_Feminist_Movement.

were sharing information about the women who had proved themselves in their careers. She expressed that in the past, American feminists challenged narrow-minded people to secure women's rights in society, and now there exists a challenge of finding human identity. She defends the fact that women also need to find their own identity, just like men, although the theorists of feminism state that women's biology is their identity. Then Friedan criticizes Sigmund Freud, the founder of psychoanalysis, because of his thoughts. Freud stated that even though women are given the rights, their position will not change. Friedan sheds light on the fact that Freud's thoughts make women who want to build a career look like neurotics, and the idea that women are only happy as housewives has become an unquestionable truth since, in that period, women were not educated enough to question and criticize his opinions. Friedan also condemned the approach of the institutions using functionalism as a part of the social sciences. According to these educational organizations, people are the social bodies of the societies, and each body should fulfill its functions; therefore, women just have to perform their biological roles as housewives and mothers, otherwise, social balance will be ruined. However, Friedan proved that this approach is wrong and unfair, taking as an example Margaret Mead, who was a famous functionalist and had a successful career as an anthropologist. In Chapter 7, Friedan focuses on how women are manipulated by education. She draws attention to the point of how the education system changed between the 1940s and early 1960s. Women were taught about family life and marriage as the ideology of functionalism had a clear impact on the educators' thoughts. They thought that teaching scientific and challenging topics to female students might ruin their femininity and prevent them from fulfilling feminine roles. Because of this wrong education system, women cannot face real-life problems and crises; unfortunately, they are stuck in the emotional phase of the growing process. Moreover, during wartime, women performed men's jobs in workplaces; however, after the war finished, women were fired, and it was claimed that soldiers who suffered from emotional disorders were raised by educated and career-oriented mothers. Friedan proves that those soldiers were raised by women who do not have a place in society as free individuals. She points out that even advertisers were against the idea of women having a career because if they work outside, they will not spend much time on house chores, so they will not buy any product to do their chores. However, critics state that those products are actually bought by men, and if women work, they will also buy such items; therefore, advertisers will make larger profits. Another considerable issue emphasized by Friedan is that children grow unhappily, with no passion or life energy because their mother cannot

live their own lives, instead they attempt their dreams to come true thanks to their children by making them perform their mothers dreams, Therefore children are neither able to realize what they want to achieve nor they can set any goal to improve themselves. In the last chapter, Friedan calls on everyone to get educated, build a career, and perform jobs that they can use their capabilities and not become stuck in monotonous house chores. This book by Betty Friedan is considered the first step of the second wave of feminism. Another successful progress achieved by Betty Friedan is the establishment of the National Organization for Women (NOW) in 1966, targeted at defending women's rights and the eradication of discrimination, injustices against women, and gender inequality.¹⁸

b) Miss America Protest (1968)

This protest was held during the contest of Miss America 1969, aiming to defend women's rights and fight against gender inequality, which played an important role in the discussion of women's roles in society. Before demonstrating a protest, they released a press pamphlet with the slogan "No More Miss America". which was prepared by Robin Morgan. The pageant was criticized for a few reasons. For example, only white women were allowed to be chosen; that's why the competition was called "Racism with Roses," and the winner was used as an advertisement for sponsors. Furthermore, women were valued just because they were quiet, beautiful, and obedient without any rejection. The participants of this protest managed to create awareness and draw people's attention to defend women's value in public. The protesters condemned the idea of objectification of women as the "beauty objects", comparing this type of competition with the cattle auction. They stated that the focus should be on their capabilities and brains. Some of the active participants are Robin Morgan, Florynce Kennedy, Bev Grant, Peggy Dobbins, Florika Remetier, Bonnie Allen, Jeannette Rankin, Kathie Sarachild, and Carol Hanisch. Symbolically, they threw the feminine things, such as high heels and makeup, into the trash can. The attendees of the protest carried slogans such as "Women's liberation", "No More Miss America", and pointed out that women suffer from limitations and discrimination in every aspect of life, including education, employment, and political fields. Every action related to this protest became the center of attention of the media. The protest accelerated the process and became an inspiration for organizing more disciplined meetings and

¹⁸Course Hero, "The Feminine Mystique Study Guide," August 11, 2017, accessed June 12, 2025, <https://www.coursehero.com/lit/The-Feminine-Mystique/>.

activism. Thanks to the protest, women's demands for equality were legitimized in public discussion. This successful protest provided a legal ground for making policy changes aiming to ban gender discrimination in education and expand reproductive rights. Women organized awareness-creating groups and were assigned to duties in political offices, which secured women's place and value in society. The rights and both men and women were given equal rights and opportunities in education, social life, and work.¹⁹

c) Gloria Steinem, "*Our Revolution Has Just Begun*"

Gloria Steinem is an American writer, political activist, and leader of feminism from the 1960s to the 1970s. In that period, she focused on two major issues: *making abortion legal and having time off from work*. These two key points made her gain fame and become a matter that was taken into consideration during the feminist movement in the US. Gloria Steinem is the founder of a few organizations supporting and defending women's rights. In her journey, Bella Abzug, Betty Friedan, Shirley Chisholm, Jane Fonda, Myrlie Evers-Williams, Robin Morgan, and hundreds of other women accompanied her. She founded organizations such as the National Women's Political Caucus (NWPC), where she delivered her speech "*Message to the Women of America*", Women's Media Center, intended for enhancing women's voices, and the magazine called Ms. in 1972, where she released her speeches and articles. One of them is "*Our Revolution has just begun*".²⁰

d) *Our Revolution Has Just Begun*

There are two misunderstandings that people ponder constantly. The first one is that people think that women's movements, all demonstrations about feminism are only for the rights of white and middle-class women, which is wrong because according to the survey made by Louis Harris and Associates in 1972 showed that the number of black women who supported the feminist activism were twice as much as white women. It is also possible to see this fact from the speeches of Shirley Chisholm, works and essays of Alice Walker, and the book of feminism "*Feminism Is For Everybody*." The second

¹⁹ Mark Engler, "The 1968 Miss America Protest - and its Significance Today," *Morningside Center for Teaching Social Responsibility*, October 14, 2018, <https://www.morningsidecenter.org/teachable-moment/lessons/1968-miss-america-protest-and-its-significance-today>.

²⁰ "Gloria Steinem," *Greenlane*, Last modified March 18, 2017, <https://www.greenlane.com/it/humanities/storia--cultura/gloria-steinem-3529174/>.

misconception is that all biases against both women and people of different races have been overcome, and there is no need to organize any movements anymore. However, this idea should not even be a matter of discussion, because even the fight for having a legal identity for all women and black men continued for more than a century, which means it will possibly take another century to ensure equality both in legal and social contexts. Today, the opposition and prejudice against women's welfare and rights, immigrants, and different races still exist, and the laws concerning these groups of people are enforced in state legislatures. Meanwhile, this country makes profits by employing women with very low wages and salaries. Unfortunately, women are not elected to political offices, and the progress in the policies in regards to child care or the solutions to family issues is not observed. Polls that are conducted in public show that today, most young women identify themselves as feminists. Then, Steinem presents the list of goals that should be achieved, each of which has the same importance. One of the matters is that "women's issues" and "economic issues" are approached as separate issues, however, the problem is women and men should be paid the same amount of wages or salaries, which would make the US economy the profit of \$200 billion and women could earn \$148 per week. This could lead to many problems being solved, for example, poor kids who are taken care of just by their mothers would be supported financially better. Besides, the career of caregiving is underestimated and under-evaluated. Caregivers are considered as the ones who do not work, however, their labor should be rewarded better because caregiving at home provides a higher quality of service and costs less than in institutions. Caregivers will make the money that they deserve when they require it. They can make this change of policy come true. Another issue is that women should be the ones who must decide when and whether they want to bear a child, unfortunately, this issue has turned to be a social issue, while it should be taken into account as a human right since it will affect the entire lives of women including their education, health and career. In addition, it is observed that women with children are underpaid, and sometimes they cannot even find a job, whereas men with children are paid well and get hired easily. The further issue, which is very critical, is that violence against women is increasing around the world. The number of women has decreased to a considerable extent. Domestic violence, teenage marriage are among the causes of this crisis. Mass shootings in public places are mainly caused by men who are obsessed with controlling others. Masculinity that is taught to children and the dominance of men in society lead to fatal crimes happening. Considering that current cultural norms provoke aggression and violence, the way of bringing up children should be redefined. In

order to make the world more peaceful and fair to everyone, the focus should be on what can be done rather than constantly thinking about what is crucial. People who have power should listen to others, and people with no power should speak up and share their thoughts and ideas, which might solve these global problems, and we should remember that there is no ranking among people, but they are connected.²¹

²¹ Gloria Steinem, “Why Our Revolution Has Just Begun”, *Herstory*, February 27, 2014, <https://msmagazine.com/2014/02/27/gloria-steinem-why-our-revolution-has-just-begun/>.

Chapter 2. Feminist Movements through NLP methods

In order to analyze the above-mentioned movements based on the works appropriately, several NLP methods were applied, such as Sentiment Analysis, Name Entity Recognition (NER), Word Cloud, and Pie Chart. Besides, some other digital tools from two different platforms, which provide the comprehensive digital analysis of data, were used: Corpus Collocates, Mandala, and Document Terms from Voyant tools, WordSketch from Sketch Engine, and the Corpus of Historical American English.

2.1. The Analysis of the First-wave of Feminism

Sentiment Analysis - As mentioned in the previous chapter, NLP is a large field that offers numerous methods and tools to analyze data. Sentiment Analysis allows us to detect the tone/emotion by evaluating the opinions in the text using scores such as positive (pos), negative (neg), neutral (neu), and compound. These categories, including positive, negative, and neutral, are scored from 0 to 1, showing from negative to positive, while the compound is evaluated based on the overall mix of other scores [-1, +1]. Two essential factors help us acquire information about the sentiment of the text: polarity and subjectivity. Polarity determines the tendency of sentiment (e.g., positive, negative, or neutral), while subjectivity shows whether the approach in the text is a subjective opinion or unbiased. Polarity is estimated in the range [-1, +1], while subjectivity is between 0 and 1. Subjectivity [0] means the opinion is objective, and [1] expresses that the information is subjective or biased. Before calculating the Sentiment Analysis for the documents with regard to the first wave of feminism, the data was preprocessed and cleaned of unnecessary parts. For example, headers and footers, stopwords, and punctuations were cleaned from the data. Here is the analysis of texts from a mathematical point of view:

$$\text{Cleaned_Text} = f_3(f_2(f_1(\text{Raw_Text})))$$

$f_1(x)$ - lowers (x)

$f_2(x)$ - removes stopwords

$f_3(x)$ - lemmatizes (x)

Then, the Textblob library was used to acquire the scores of polarity and subjectivity for each document separately. Here is the statistical explanation of the concepts:

Polarity:

$$Polarity_{total} = \sum_{i=1}^n w_i \cdot p_i / \sum_{i=1}^n w_i$$

n - the number of words with sentiment

w_i - weight assigned to each word based on intensifiers, negation, punctuation, and capitalization

p_i - the base sentiment score of the word, ranging from -1 to +1

$\sum_{i=1}^n$ - sigma notation: sums the expression $w_i \cdot p_i$

Subjectivity:

1.0 - Opinion/message is completely subjective

0.0 - opinion/message is completely objective

$$Subjectivity_{total} = 1/n \sum_{i=1}^n s_i$$

n - the number of words with subjective meaning

s_i - the subjective score of the i^{th} word

Σ - 'sigma' to sum the scores for all n words

$1/n$ - Divide the total by the number of words to find the average

Here are the Polarity and Subjectivity scores for each document regarding the first wave of feminism:

	Polarity	Subjectivity
Seneca Falls Convention	0.1391	0.4561

Address on Women's Rights	0.1414	0.5153
Trial of Susan B. Anthony	0.0859	0.4301

Overall, all documents are slightly positive in terms of polarity, ranging approximately from 0.08 to 0.14, while they show average scores for subjectivity, meaning that the data in documents is neither biased nor subjective. This neutrality in the subjective scores proves that they demand equality for both genders based on objective and balanced requirements without insulting any specific group of people.

Negative: $neg = \frac{\sum_{v_i \in N} |v_i|}{(\sum_{v_i \in P} v_i + \sum_{v_i \in N} |v_i| + Z)}$

Neutral: $neu = Z / (\sum_{v_i \in P} v_i + \sum_{v_i \in N} |v_i| + Z)$

Positive: $pos = \frac{\sum_{v_i \in P} v_i}{(\sum_{v_i \in P} v_i + \sum_{v_i \in N} |v_i| + Z)}$

Compound $compound = \tanh(\sum v_i / \sqrt{\sum v_i^2 + \alpha})$

v - the value of the i^{th} word

N - the set of negative value scores

P - the set of positive value scores

Z - the number of neutral words

α - a constant to normalize the process of scaling the compound score

The following results demonstrate the scores for each document:

	Negative	Neutral	Positive	Compound
Seneca Falls Convention	0.127	0.666	0.207	0.9988
Elizabeth Cady Stanton's Address on Women's Rights	0.131	0.6	0.269	1.0
Trial of Susan B. Anthony	0.11	0.74	0.15	1.0

As shown, the documents are mainly neutral, and the negativity is less than positivity, while the Compound scores are highly positive, which shows that overall sentiment is positive.

After receiving the general information, each document was analyzed at the sentence level, and the purpose was to figure out whether the message delivered in each sentence is positive, negative, or neutral. The results for each file are below:

Seneca Falls Convention:

	sentence	sentiment
0	The first convention ever called to discuss th...	positive
1	Woman's rights convention\nTHE FIRST CONVENTIO...	positive
2	Section VII No.1 No.16\nWOMAN'S RIGHTS CONVENT...	neutral
3	A Convention to discuss the social, civil, and...	positive
4	The public generally are invited to be present...	negative
..	...	...
89	In every generation\nGod calls some men and wo...	positive
90	But we will steadfastly abide the result.	positive
91	Unmoved we will bear it aloft.	neutral
92	Undauntedly we will\nunfurl it to the gale, fo...	positive

Elizabeth Cady Stanton, “Address on Women’s Rights”:

	sentence	sentiment
0	ELIZABETH CADY STANTON, “ADDRESS ON WOMAN’S RI...	negative
1	Should that gentleman be present this evening ...	positive
2	I should feel exceedingly diffident to appear ...	positive
3	Man cannot speak for us-because\nhe has been e...	positive
4	Moral beings can only judge others by\nthemsel...	positive
..	...	...
281	will ye look to that\nwhich cannot satisfy you.	neutral
282	Will you waste your time and strength on lower...	negative
283	Will you reverse that, that is of like nature ...	neutral
284	Then fear not thou to wind thy horn,\nThough e...	neutral
285	Ask for the castles King and Queen,\nThough ra...	negative

The Trial of Susan B. Anthony:

		sentence	sentiment
0	The Trial of Susan B. Anthony\nby\nAnn D. Gord...		positive
1	Circuit Court for the Northern District of New...		positive
2	15\nDid the Fourteenth and Fifteenth Amendment...		neutral
3	15\nDid women have voting rights under the Con...		neutral
4	16\nCould a belief in a right to vote be used ...		positive
...		...	...
1242	This research assists Judicial Conference comm...		positive
1243	The Center's research also\ncontributes substa...		positive
1244	The two divisions work closely with two units\...		negative
1245	The Federal Judicial History Office helps\ncou...		neutral
1246	The International Judicial Relations\nOffice p...		negative

As can be seen, each document contains sentences with all three labels depending on the number of words with bad and good meanings. It should be remembered that sometimes, mean words may be used more than the good ones to express an opinion that is not offensive and aims to deliver a hidden or moral message. For example, Elizabeth Stanton asks in her writing: “Will you waste your time and strength on lowering buckets into empty wells?” which was estimated as a negative opinion by Sentiment Analysis coding, however there is a message in this sentence that says not to waste time or strength on things that there is no benefit in. This displays that Sentiment Analysis cannot distinguish figurative language. The sentence “Now is the time for the women of this country, if they would save our free institutions, to defend the right, to buckle on the armor that can best resist the keenest weapons of the enemy — contempt and ridicule.” from the document Seneca Falls Convention was estimated Neutral since no offensive message was detected.

Word Cloud

Another way to get information from data is to print their word clouds to see what type of contexts were discussed during that period. In order to achieve the expected results, the Matplotlib library was applied.

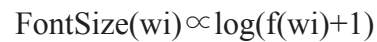
The structure of cloud computing:

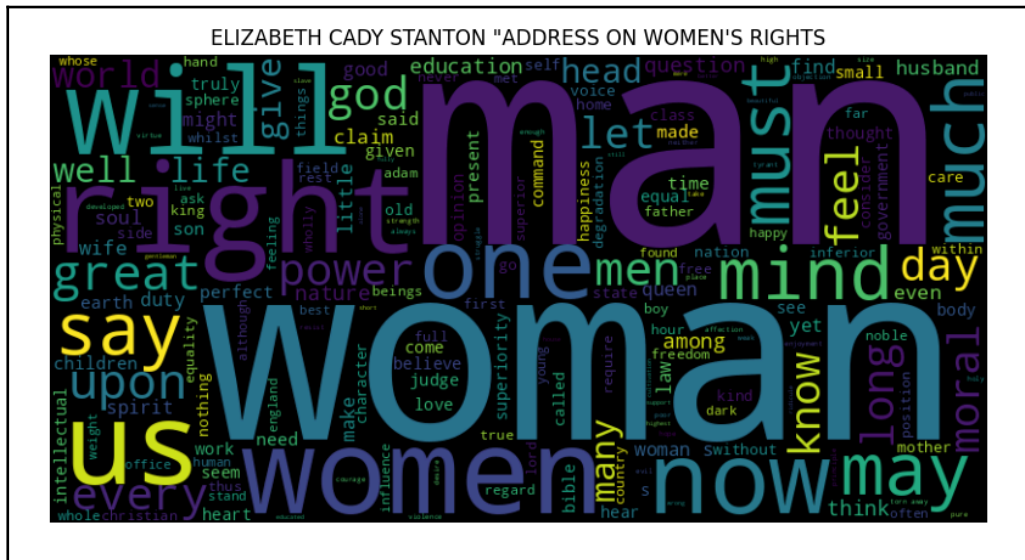
Each file becomes a sequence of words:

$$D_k = \{wk_1, wk_2, \dots, wkn_k\}$$

$$D = \cup_{k=1}^N D_k = \{w_1, w_2, \dots, w_3\}$$

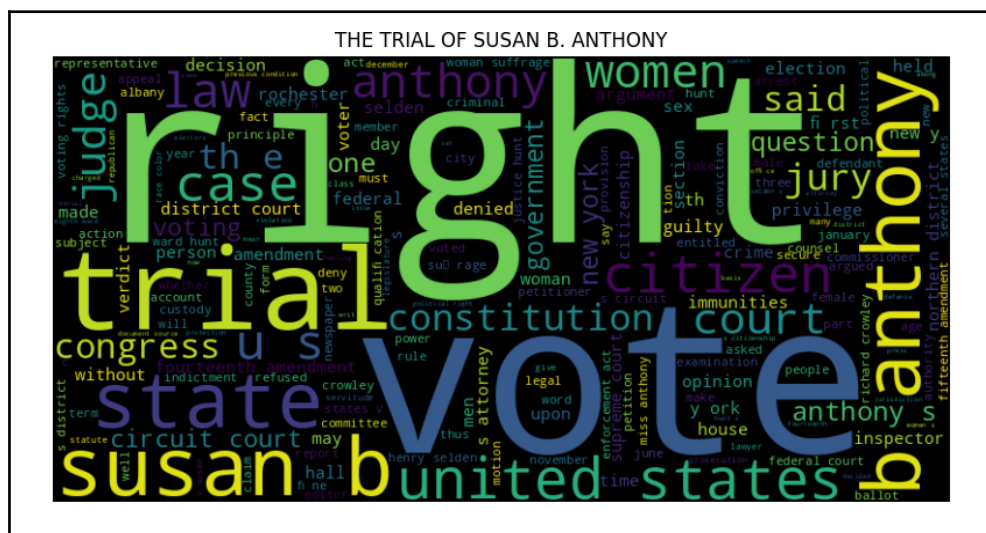
$$n = \sum_{k=1}^N n_k$$

$$f(w_i) = \text{count of } w_i \text{ in } D$$
$$f(w_i) = \#(w_i) = \sum_j^n \delta(w_i w_j)$$
$$w_i = w_j, \delta(w_i w_j) = 1$$
$$\delta(w_i w_j) = 0$$
$$\text{FontSize}(w_i) \propto \log(f(w_i)+1)$$
$$\text{FontSize}(w_i) \propto \log(f(w_i)+1)$$
$$\text{FontSize}(w_i) \propto \log(f(w_i)+1)$$

$$\text{FontSize}(w_i) \propto \log(f(w_i)+1)$$
$$\text{FontSize}(w_i) \propto \log(f(w_i)+1)$$



Here in this context, the words – will, right, man, woman, mind – are the ones that deliver the message of her writing.

Susan B. Anthony was judged for voting, therefore, her trial document printed a law-oriented output:



As can be seen, this document contains keywords such as trial, state, right, vote, constitution, court, case, and judge, as it pertains to a judicial event at the law level. The primary difference from the previous documents is the official judgment by the law, whereas others were intended to discuss and balance rights.

Named Entity Recognition (NER) -

Another method that was already mentioned is to apply named entity recognition (NER) so that named entities can be observed. For this method, the spaCy library was used. SpaCy is such an efficient library that allows us to analyse data comprehensively. With spaCy, it is possible to work on data for a few tasks such as tokenization, part-of-speech tagging, named entity recognition, dependency parsing, lemmatization, text classification, deep learning tasks, and so on. For the documents related to the first and second waves of feminism, spaCy was applied to print named entities, which enables us to see who participated in those events. The structure of NER computation:

A string of tokens for each document:	$D = \{w_1, w_2, \dots, w_n\}$
Application of spaCy to D:	$Doc = nlp(D)$
The extraction of NEs:	$Doc.ents = \{e_1, e_2, \dots, e_k\}$
e_j = a combination of a token and its entity	$e_j = (s_j, l_j)$
s_j = token	
l_j = the entity of the token	

Here are the outputs for each document separately:

Seneca Falls Convention:

If we pay attention to the label PERSON, we can see such names: Lucretia Mott, Martha Wright, Mary Ann McClintock, Elizabeth Stanton, Charlotte Woodward, Richard P., Hunt Samuel, James Mott, Thomas McClintock, and Joan Arc attended the conference. Here we observe the male names, which show that there were also men who supported gender equality.	<pre> --- Named PERSON Entities in Document 1 --- lucretia mott philadelphia lady (PERSON) - 480:511 martha (PERSON) - 687:693 wright mary ann mcclintock jane (PERSON) - 696:727 mott elizabeth (PERSON) - 4122:4136 mary ann mcclintock (PERSON) - 4170:4189 martha (PERSON) - 4190:4196 wright jane (PERSON) - 4199:4210 charlotte woodward sarah (PERSON) - 4264:4288 richard p (PERSON) - 4299:4308 hunt samuel (PERSON) - 4309:4320 james mott (PERSON) - 4373:4383 thomas mcclintock (PERSON) - 4384:4401 joan arc (PERSON) - 8805:8813 joel ii (PERSON) - 9178:9185 </pre>
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Elizabeth Cady Stanton’s Address on Women’s Rights:

The list of persons shows that the number of men is higher than that in the conference, e.g., John Quincy, Daniel Webster, Gerrit Smith, Peter the Great, Joan of Arc, etc. Stanton states that John Quincy Adams, the 6th President of the U.S., was a small man with with high intellect, which shows that superiority must not be measured based on physical appearance but intellectual capabilities. Based on this fact, women should also be treated equally since they are highly intellectual human beings. She pointed out that prominent people of the period, such as Senator Daniel Webster and abolitionist Gerrit Smith, defended the rights of those who were uneducated or unqualified, but ignored the capabilities of women.	--- Named PERSON Entities in Document 2 --- elizabeth (PERSON) – 0:9 de stael (PERSON) – 3697:3705 herschel mary summerville scientific (PERSON) – 3735:3771 john quincy (PERSON) – 9757:9768 barrister lord mayors (PERSON) – 11892:11913 daniel webster (PERSON) – 13021:13035 webster van buren (PERSON) – 13508:13525 gerrit smith (PERSON) – 13531:13543 son (PERSON) – 17772:17775 paul (PERSON) – 18141:18145 meek (PERSON) – 20780:20784 margaret (PERSON) – 23447:23455 elizabeth (PERSON) – 24332:24341 james catharine (PERSON) – 24348:24363 peter great (PERSON) – 24398:24409 hannah (PERSON) – 24965:24971 christopher columbus dark hour (PERSON) – 25610:25640 joan arc (PERSON) – 28224:28232 castle king (PERSON) – 30264:30275
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The Trial of Susan B. Anthony:

In this document, the majority of people is male, because it is about the trial of Susan B. Anthony regarding her vote in the election, and a lot of men were against women's rights and gender equality. Here are the female participants: Ann Gordon, Elizabeth Cady Stanton, Susan B. Anthony, etc. Male participants: William C. Storrs, Richard Crowley, Henry Selden, John Bingham, John Van, Matthew Carpenter, Edwin Marsh, James Wasson, James McGinty, Henry Davis, James Kenney, Peter H. Bradt and so on.	<pre> --- Named PERSON Entities in Document 3 --- ann gordon (PERSON) - 0:10 elizabeth cady stanton (PERSON) - 25:47 anthony short (PERSON) - 552:565 susan b anthony lawyers (PERSON) - 1405:1428 william c storrs (PERSON) - 1484:1500 susan b anthony indict voter (PERSON) - 1543:1571 john van (PERSON) - 1642:1650 richard crowley (PERSON) - 1659:1674 john e pound medium (PERSON) - 1675:1694 anthony (PERSON) - 1834:1841 susan b (PERSON) - 1853:1860 anthony henry (PERSON) - 1861:1874 richard crowley (PERSON) - 1922:1937 susan b anthony speech (PERSON) - 1949:1971 susan b anthony elizabeth (PERSON) - 2160:2185 anthony speech (PERSON) - 2243:2257 anthony bibliography (PERSON) - 2371:2391 anthony criminal (PERSON) - 2486:2502 anthony (PERSON) - 3151:3158 anthony guilt (PERSON) - 3422:3435 anthony short (PERSON) - 3732:3745 anthony criminal (PERSON) - 3793:3809 anthony (PERSON) - 3856:3863 anthony (PERSON) - 4374:4381 anthony (PERSON) - 4954:4961 anthony (PERSON) - 5174:5181 anthony quali (PERSON) - 5332:5345 anthony oath (PERSON) - 5428:5440 anthony crime (PERSON) - 5597:5610 strategy susan (PERSON) - 6006:6020 anthony expect (PERSON) - 6023:6037 anthony members association (PERSON) - 6343:6370 william storrs (PERSON) - 7043:7057 anthony (PERSON) - 7098:7105 henry selden (PERSON) - 7560:7572 john pound pound (PERSON) - 7594:7610 anthony plea selden (PERSON) - 7615:7634 anthony action (PERSON) - 7812:7826 anthony (PERSON) - 7887:7894 </pre>
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Voyant tool ‘CorpusCollocates’²²

In addition to the research in Python, other platforms are helpful in terms of analyzing data from the aspect of computational linguistics, which is called Voyant tools²³. This is the platform in which academic tasks can be performed through diverse methods, for instance, concerning the visualization of term frequency, the methods like BubbleLines, Bubbles, and Document Terms can be used, or the CorpusCollocates method identifies which words a term has been used in the text. Besides, it is possible to get word clouds with the Cirrus method and can also examine the context by studying collocations through the tools such as Documents, Reader, Summary, and Topics. CorpusCollocates method analyses data by taking into account the word, its collocates, and window size, and uses statistical association measures. Pointwise Mutual Information (PMI) is a measure to identify how often two words appear together. PMI is usually used in information retrieval, corpus linguistics, and NLP tools such as Voyant.

$$PMI(w, c) = \log_2((P(w, c)/P(w).P(c))$$

²² <https://voyant-tools.org/docs/tutorial-corpuscollocates.html>

²³ https://voyant-tools.org/docs/tutorial-tools_.html

P - probability

w - word

c - a collocate

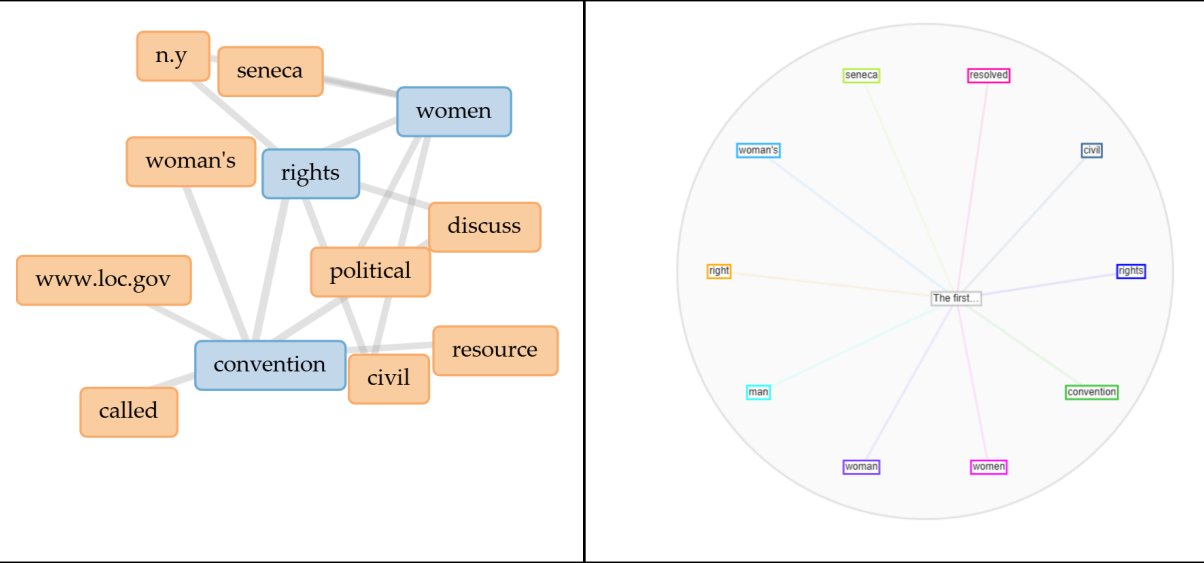
The Analysis of **Seneca Falls Convention via CorpusCollocates**, which shows the words with ones that they are used together: The words have collocates with the different meanings such as severely, sad, objection, object, fallen, degradation, dark, tyranny, injurious, inferior and so on, which describe the life women have been going through.

	Collocate	Frequency
rights	deprived seneca falls equal exercise secure duty	1 8 8 2 2 1 2
women	inferior, severity, oppressed, onlytolerated, occupied, aggrieved	1
man	tyranny, scorn, object, inferior, injurious, abuse, wrong, suffer, severity	1

Voyant tool Collocates Graph²⁴ & ‘Mandala’²⁵- Collocates Graph and Mandala are also two other tools that allow us to analyze the keywords in connection with their collocates. According to the current results, women were deprived of their civil rights; therefore, the conference aimed to put an end to this ongoing unfair treatment.

²⁴ <https://voyant-tools.org/docs/tutorial-collocatesgraph.html>

²⁵ <https://voyant-tools.org/docs/tutorial-mandala.html>



Voyant tool ‘Document Terms’²⁶ - Based on the frequencies of the terms, the Conference focused on resolving the issues related to political positions of women in society by requiring the provision of their rights.

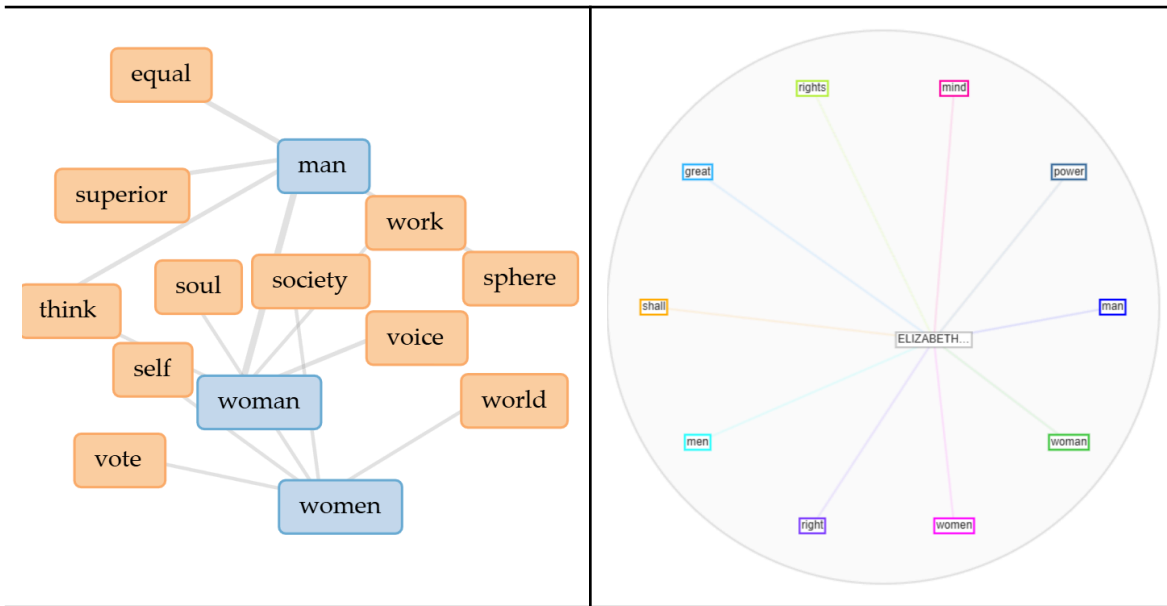
	Count	Relativity
rights	28	9,530
convention	22	7,488
women	21	7,148
woman	14	4,765
right	13	4,425
resolved	11	3,744
seneca	11	3,744
laws	9	3,063
political	9	3,063
position	6	2,042
work	4	1,361

²⁶ <https://voyant-tools.org/docs/tutorial-documentterms.html>

Elizabeth Cady Stanton's Address on Women's Rights - Here the word 'man' is linked to words such as unjust, unfit, tyranny, suffering, suffered, disgraceful, destroy, denial, contention, chastise, etc., in the following tables:

	Collocate	Frequency
right	violate, imprison, deny chastise, wrong	1
	woman, vote, claim	2
woman	degradation, crushed, crisis, absurd	1
man	superior	3
	right, support	2
	hates, chastise, unjust unfit, tyranny suffering, scorn, poor	1

Voyant tool Collocates Graph & 'Mandala'- Stanton addressed social rights of women, highlighting that women's voice should be heard, so that they can receive fair treatment by getting a proper education and job, voting in elections, and playing a role in society. The term 'superior' is directly related to 'man', which shows that she criticizes the concept of superiority that women were made to think that men deserve such treatment.



Document Terms - Likewise, Document Terms also provides identical results. However, as shown, the domestic life of women was discussed by Stanton. The terms such as wife, children, education, and moral demonstrate that the society attempts to impose such a perception on women that they are expected to be a wife and raise their children, which was supposed to be their only job.

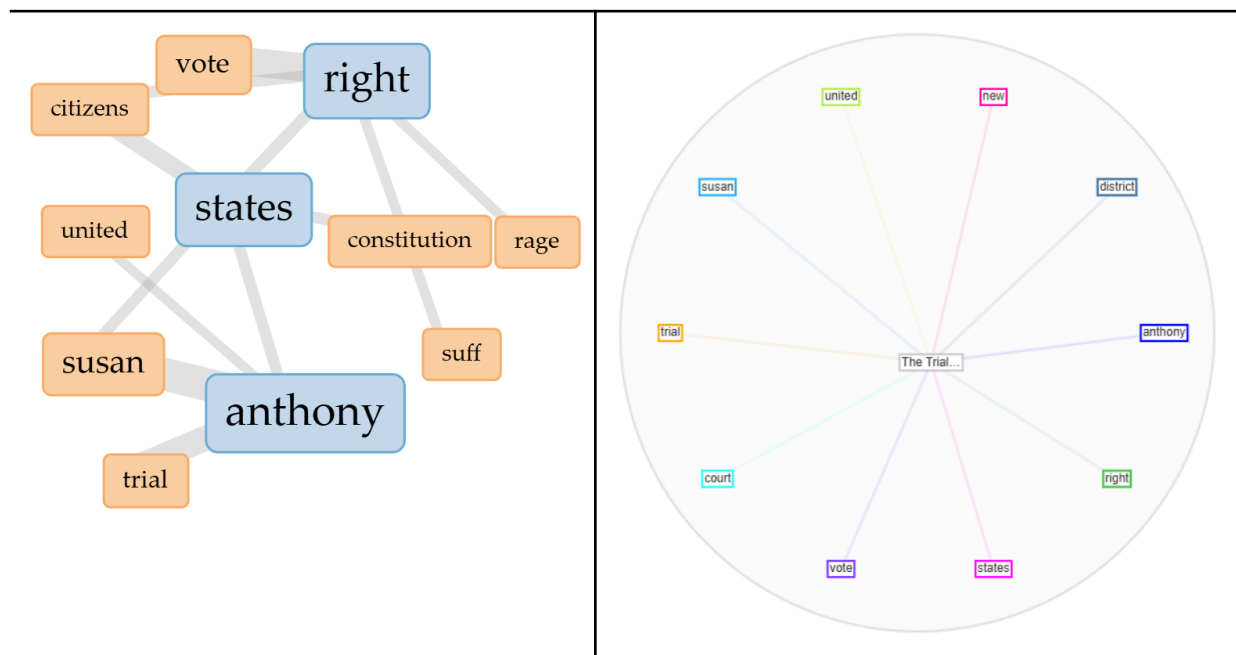
	Count	Relativity
man	65	6,974
woman	46	4,935
women	38	4,077
right	24	2,575
men	22	2,360
rights	19	2,038
mind	18	1,931
power	17	1,824
moral	12	1,287
education	12	1,287

wife	11	1,180
laws	9	966
children	9	966

The Trial of Susan B. Anthony - This document contains negative expressions, for example, violation, illegal, deprived, criminal, prisoner, etc., since Susan B. Anthony was accused of voting, which was considered illegal:

	Collocate	Frequency
right	rage	18
	deny	11
	wrong	4
	violation, crime, unlawful, restrict	2
	refuse, restriction, sue, struggle, unlawful, neglected	1
vote	crime	3
	refuse, lacked, doubt, deny	2
anthony	criminal	7
	guilty, crime	5
	violation, refused, illegal, denied	3
	miss, lacked, conceded	2
	violate, unjust, prisoner, setbacks, penalty, deterrent	1
states	deny	7
	restrict	3
	rage, failing	2

Voyant tool ‘Mandala’ - It is obvious from this trial that voting was a serious crime for women in that period, which resulted in the payment of a fine that Anthony refused to pay, claiming that, as citizens of the U.S., women have the right to vote. This case proves that women were exposed to many unfair treatments by men, which have no sensible explanation for any of them.



	Count	Relativity
right	295	9,550
states	285	8,166
vote	251	6,948
court	233	6,449
trial	232	6,422
united	182	5,038
rights	122	3,377
women	119	3,294
amendement	112	3,100

case	113	3,128
federal	111	3,072
voting	104	2,879
law	102	2,823
justice	52	1,439
crime	51	1,412
criminal	49	1,356
privileges	47	1,301
verdict	44	1,218
guilty	44	1,218
suffrage	43	1,190
men	41	1,135
voters	40	1,107
citizen	40	1,107
argument	40	1,107
constitution	90	2,491
congress	84	2,346

SketchEngine ‘WordSketch’

SketchEngine²⁷ is another platform that helps to analyze data under a large scope, including the use of WordSketch, Concordance, Thesaurus, the extraction of Wordlist, and so on. WordSketch is a method that shows the words in their context, using statistical measures such as logDice, measuring how strong the relation is between the word and its collocate, PMI, and T-Score, which is used to ignore unnecessary associations.

²⁷ <https://www.sketchengine.eu/>

$$\logDice(w, c) = 14 + \log_2((2 * f(w, c)/f(w) + f(c))$$

$f(w, c)$ - The frequencies of w and c together

$f(w), f(c)$ - The individual frequencies of w and c

$$T - score(w, c) = f(w, c) - E(w, c) / \sqrt{f(w, c)}$$

$f(w, c)$ - detected co-occurrence frequency

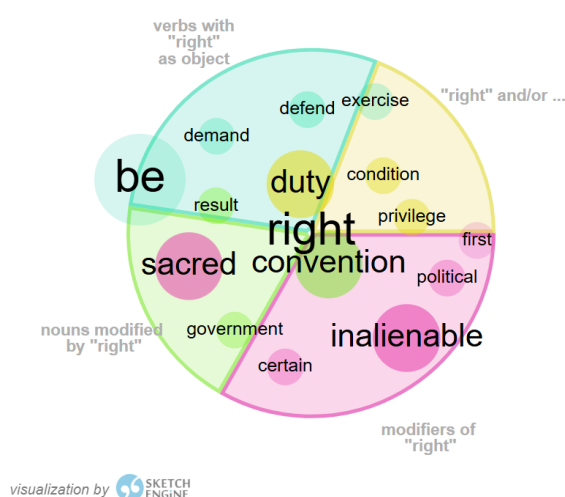
$$E(w, c) = f(w) * f(c) / N$$

N is the total number of tokens in the corpus

Seneca Falls Convention - The word 'right' is used together with expressions like *rights convention, rights government, defend the right, demand the right, and exercise their inalienable right*, which shows that in these documents, the rights were demanded to exercise women's rights. Here, the expression '*exercise her inalienable right*' shows that the main topic is about the requirement of women's rights.

modifiers of "right"

sacred
sacred rights
inalienable
inalienable rights
certain
certain inalienable rights
political
political rights
first
first right



nouns modified by "right"

convention
rights convention
result
rights results
government
rights governments

verbs with "right" as object

defend
defend the right
demand
demanding the right
exercise
exercise her inalienable right
be
is the right

"right" and/or ...

duty
right and duty
privilege
rights and privileges
condition
condition and rights

Elizabeth Cady Stanton 'Address on Women's Rights' - The writing of Elizabeth Cady Stanton contains law-related terms such as *equal right*, *discuss our rights*, *declare our rights and privilege*, *claim the right interest*, *secure this right and justice*, which means that Stanton has focused on providing women with the rights that they deserve.

modifiers of "right"

civil
has all the civil rights
government-our
government-our rights
equal
equal right
same
same rights



verbs with "right" as object

declare
declare our right
have
has no right
secure
secure this right
man
man the right
discuss
discuss our rights
claim
claim the right

possessors of "right"

Woman
Woman's rights
woman
women's rights

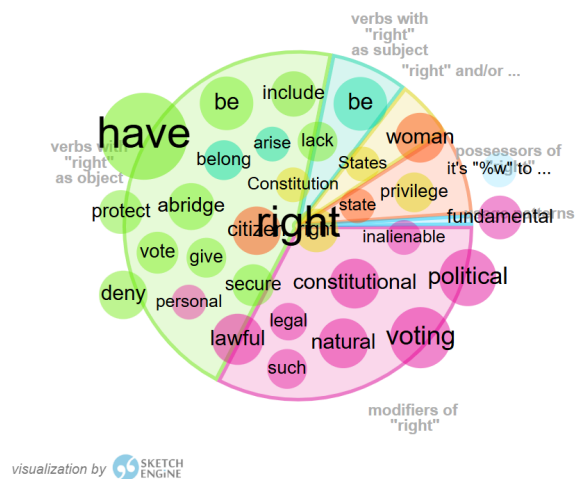
"right" and/or ...

privilege
rights and privileges
interest
rights and interests
wrong
rights and wrongs
justice
right and justice

The Trial of Susan B. Anthony - Susan B. Anthony was found guilty in court because her vote in the election was considered illegal. However, she claimed that women have the right to vote since they are also citizens. Therefore this document contains the key phrases such as *constitutional, voting, lawful, political, vote, extend voting rights, give the rights, confer the right, inalienable right, secure the most valuable rights, personal rights, natural right, the fundamental rights, citizen's right to vote, women's right to vote.*

modifiers of "right"

voting	noun
voting rights	
political	
political rights	
lawful	
without having a lawful right to vote in	
natural	
a natural right	
constitutional	
constitutional right to vote	
fundamental	
that the fundamental rights of	
such	
that she had such right	
legal	
legal right to vote	
personal	
personal rights	
inalienable	
inalienable right	
voting	adjective
Did women have voting rights under the Constitution	
human	
human rights	



verbs with "right" as object

have
had a right to vote
deny
deny the right to
abridge
abridging the right
include
include the right of
be
was a natural right
protect
protected the right of women to
secure
secure the most valuable rights
vote
voting rights to
lack
she lacked the right to vote
give
gives the right
confer
confer the right
extend
extend voting rights

verbs with "right" as subject

belong
a right belonging to
arise
that voting rights arose from state citizenship
be
right is

possessors of "right"

citizen
citizen's right to vote
woman
women's right to vote
state
states' right

"right" and/or ...

right
My natural rights , my civil rights , my political
privilege
right or privilege
Constitution
Voting rights and the Constitution
States
States , the right to

pronominal possessors of "right"

my
the denial of my right
their
their right to vote
her
her right to vote
his
his personal rights

Pie Chart

The steps of computing a Pie Chart:

Counting words:

$$\sum_i \delta(w_i, \text{word})$$

Frequency:

$$\text{Vector } \vec{f} = [f_1, f_2, \dots, f_k]$$

Angle of Pie slice:

$$\theta_j = f_j / \sum_i f_i \cdot 360^\circ$$

Percentage of Pie slice:

$$p_j = f_j / \sum_i f_i \cdot 100\%$$

As a final phase of analyzing the first wave of feminism in U.S. history, we can examine the pie charts of each document according to their most commonly used:



2.2. The Analysis of the Second-wave of Feminism

The second wave of the feminism movement - The same methods have been applied to the documents of the second wave of feminism. As already mentioned, the feminist movement of this period focused on determining and maintaining women’s rights and balancing gender and social equality in the different fields of society, including domestic, career, and social life. Now, let us examine the results obtained from each method:

Sentiment Analysis of The Feminine Mystique by Betty Friedan (1963) - As can be seen from below, the negative and positive scores are low, while that of neutral is above average. This shows that the documents are mainly neutral, however, the compound category has extreme scores. The first document's compound score is 1.0, and the third one's is 0.9797, which means that the overall sentiment of these two documents is highly positive, while the compound score of the second document is extremely negative, displaying a very low score: -0.8159.

	Negative	Neutral	Positive	Compound
The Feminine Mystique	0.127	0.688	0.185	1.0
The Miss America Protest	0.171	0.687	0.142	-0.8159
Our Revolution Has Just Begun	0.138	0.689	0.173	0.9797

Another way to acquire general information about the files is to look through the calculations of polarity and subjectivity. Based on the following results, it is possible to say that the overall language of all documents is slightly positive, and the opinions and ideas are relatively subjective.

	Polarity	Subjectivity
The Feminine Mystique	0.0927	0.4505
The Miss America Protest	0.1150	0.3739
Our Revolution Has Just Begun	0.0458	0.3753

Here are the general results of each document, and a few example sentences from the files:

The Feminine Mystique by Betty Friedan -

		sentence	sentiment
0	Further praise for The Feminine Mystique\n“Wr...		negative
1	That The\nFeminine Mystique is at the same tim...		positive
2	Betty Friedan is to women what Martin Luther K...		positive
3	Things\nare different between men and women be...		negative
4	Betty gave them to us.”\n–Caroline Bird, autho...		positive
...		...	...
6198	What is the cause of their\nanguish?		neutral
6199	Is society forcing women to be unhappy?		negative
6200	To what\nextent are these women responsible fo...		positive
6201		7.	neutral
6202	“Perhaps it is only a sick society, unwilling ...		negative

The evaluation result of the third example sentence is neutral because no observation of positivity or negativity was made.

File: The Feminine Mystique.pdf
Sentence: This questionnaire was later used by alumnae classes of Radcliffe and other women’s colleges with similar results.
Sentiment Label: Neutral

Miss America Protest -

	sentence	sentiment
0	The 1968 Miss America Protest - and its Signif...	positive
1	"Everybody tuned into Miss America back then-t...	neutral
2	Making\nnational headlines, the protest thrust...	neutral
3	Writer and professor Roxane Gay\ndescribed the...	positive
4	They would "protest the image of Miss\nAmerica...	negative
5	Male reporters would not be\nallowed to interv...	neutral
6	The organizers obtained a permit, detailing th...	negative
7	Protesters held signs\nwith such statements as...	positive
8	One woman performed a\nskit, holding her child...	neutral
9	A prominent black feminist activist and\nlawye...	negative
10	"The Atlantic City action,"\nKennedy continued...	negative
11	The Man never expects the second kind of prote...	positive
12	It was a compelling\nimage...	positive
13	But it never actually happened.	neutral
14	In fact, officials asked the women not\nto set...	neutral
15	The 1968 protest connected the oppression of w...	positive
16	As historian Paige Welch explained in a\nSepte...	positive
17	Less well known is that they saw the pageant a...	positive
18	The organizers had roots in radical leftist ca...	negative
19	Upon descending on Atlantic City, women's libe...	negative
20	After the event, protest organizers celebrated...	positive
21	In a 1968 essay, organizer Carol Hanisch noted...	positive
22	Later writing in The Feminist Memoir Project, ...	positive
23	They no longer felt so alone and Isolated.	neutral
24	In subsequent years, the feminist movement sco...	positive

The next sentence from the second document shows the label 'negative', since it contains the feeling of anger and the provocation to fight for fair treatment.

File: The 1968 Miss America Protest - and its Significance Today.pdf
Sentence: Protesters held signs
with such statements as "All Women Are Beautiful," "Cattle parades are
demeaning to human beings," "Don't be a play boy accessory," "Can make-up
hide the wounds of our oppression?"
The protesters adopted guerrilla theater tactics, too.
Sentiment Label: Negative

Gloria Steinem: *Our Revolution Has Just Begun*

		sentence	sentiment
0	Our Revolution Has Just Begun\nPUBLISHED 3/21/...		positive
1	Join us each\nday as we bring you the stories ...		positive
2	The following is excerpted from a talk given b...		neutral
3	The full text\nappears in the Winter/Spring 20...		positive
4	If I had to pick a couple of myths about the w...		negative
..		...	...
05	And remember, the end doesn't justify the mean...		neutral
06	If we\nwant dancing and laughter and friendshi...		neutral
07	That is the small and the big of it.		negative
08	At my age, in this still hierarchical time, pe...		positive
09	Because only if each of us has a torch will th...		positive

The following sentence is evaluated with the label 'positive'. The sentence contains a positive and encouraging message, and this is possibly the reason for the positive result.

```
File: Our Revolution Has Just Begun.pdf
Sentence: Join us each
day as we bring you the stories of iconic and soon-to-be-famous feminist
change-makers.
Sentiment Label: Positive
```

The next method is the acquisition of word clouds:

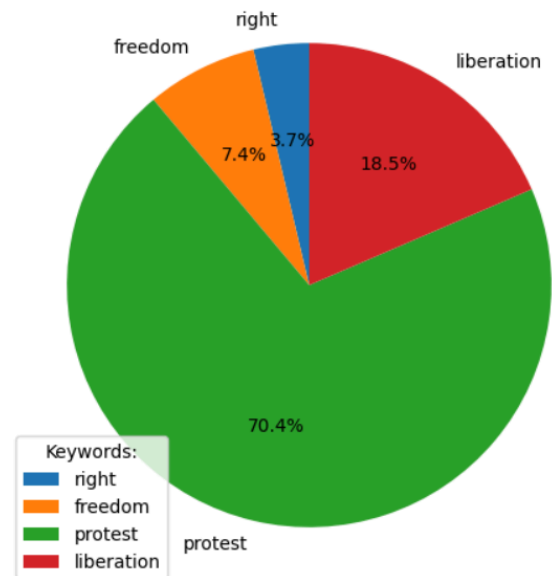
The Feminine Mystique by Betty Friedan - As displayed here, the main topics discussed in this file are about domestic and work life, women's role in social life, education, marriage, college, housework, etc.



Miss America Protest -

Supporters of this protest demanded the freedom and liberation of women from being objectified and competing with each other for their appearance, saying that all women are beautiful, with posters.

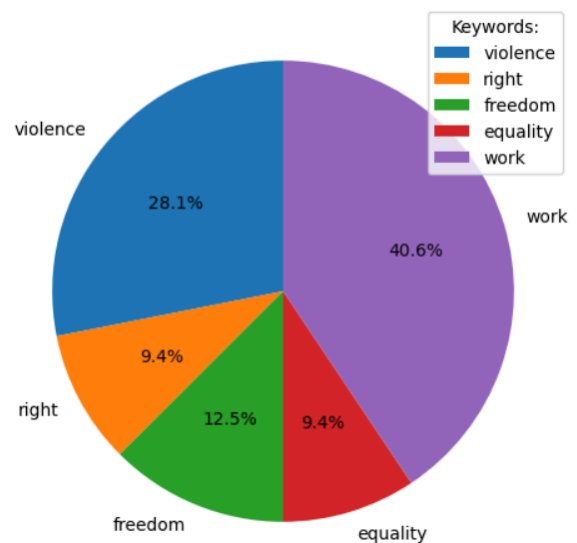
Word Frequency in The 1968 Miss America Protest



Gloria Steinem 'Our Revolution Has Just Begun'

Gloria Steinem stated in her speech that women should build their careers and learn how to stand on their foot. In this pie chart, it seems that she talked about a wide range of topics elaborately, focusing more on work and violence.

Word Frequency in 'Our Revolution Has Just Begun'



Obtaining named entities from the documents will allow us to see participants and supporters of the second wave of feminism.

The Feminine Mystique by Betty Friedan -

All these people played a supportive role in the improvement of the feminist movement by their writings and studies later on. For example, Dr. Margaret Mead and Ruth Benedict, who are well-known anthropologists, conducted culture and gender-related research, which unveiled women's rights and roles in society. David Riesman, who was a sociologist who wrote a book called 'The Lonely Crowd', mentioned how women are manipulated and confined by social pressures that suppress their ambitions and aim to make them just as housewives. Dr. Margaret Lawrence was a psychiatrist focused on female psychology and the condition of women's mental health regarding their domestic life. Dr. Lois Meek was another psychologist who conducted medical studies on female psychology. Norman Mailer and Tennessee Williams are critics of that period, who focused on gender roles and societal expectations in America.	sarah lawrence (PERSON) margaret (PERSON) lie letter (PERSON) laurel crown genius (PERSON) einstein schweitzer roosevelt edison (PERSON) dr james (PERSON) david riesman (PERSON) margaret (PERSON) marry catholic (PERSON) ego superego (PERSON) dr margaret lawrence (PERSON) conscience stricken require (PERSON) julie graduate (PERSON) moon (PERSON) billboard billboard (PERSON) max (PERSON) jacob (PERSON) margaret (PERSON) bull (PERSON) marry twenty (PERSON) matinee (PERSON) mccarthy uncontrollle (PERSON) norman mailer (PERSON) tennessee williams (PERSON)
--	---

Miss America Protest

Alix Kate Shulman, one of the protestants, stated that being chosen Miss America and winning an Oscar had the same value for people. Florence Kennedy was a black lawyer and feminist who chained herself to a doll of Miss America to demonstrate the current treatment that women were exposed to – enslavement of women by beauty standards. Robin Morgan was one of the organizers of the protest and continued her activities in the improvement of the feminist movement.	alix kate (PERSON) kennedy (PERSON) latina native american (PERSON)
--	--

Gloria Steinem ‘Our Revolution Has Just Begun’ -

Louis Harris was an analyst and journalist who founded the polling firm Louis Harris & Associates, surveying the challenges against gender equality and the status of women in society. According to the survey, this movement was also supported by black women. Steinem referred to this survey during her speech to express that this movement was not just for white women; on the contrary, it was meant for all women from different races and classes. Shirley Chisholm was a black American politician and educator who ran for President in 1972. Steinem showed Chisholm as an example to highlight that gender equality is also connected to racial justice, and no matter	gloria steinem march (PERSON) gloria steinem (PERSON) grrrls white (PERSON) louis harris associates (PERSON) shirley chisholm (PERSON) dorothy dinnerstein (PERSON) u murder (PERSON) valerie hudson (PERSON) philip roth (PERSON)
--	---

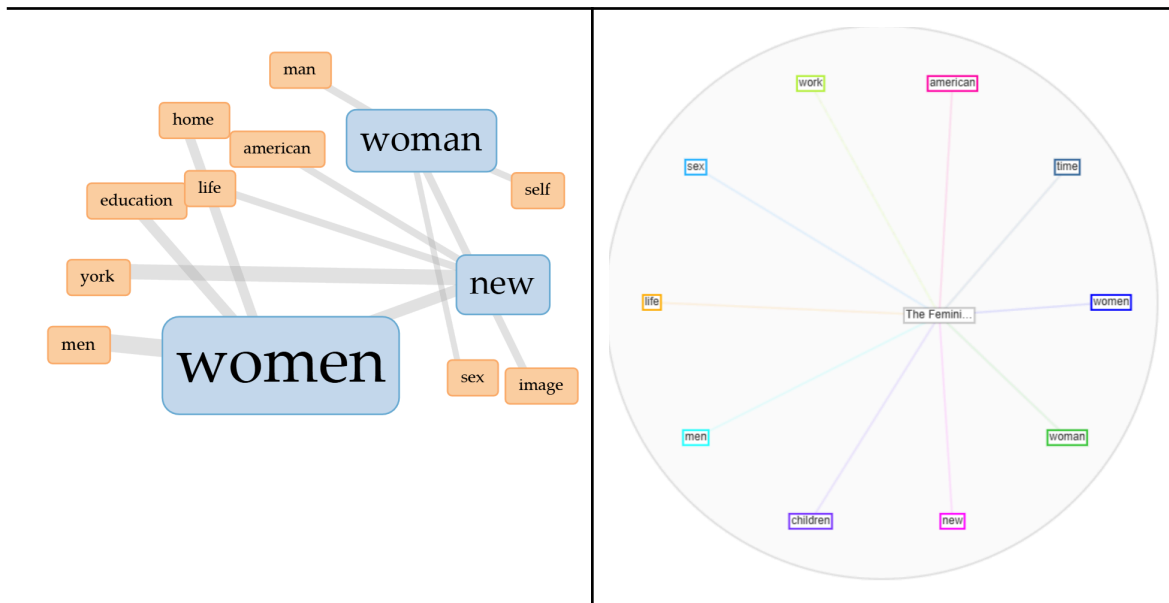
what her color is, all women should fight to win the rights they deserve. Dorothy Dinnerstein was a feminist and psychologist who talked about gender roles and power from a psychological aspect, which influenced Steinem. Based on Dinnerstein's viewpoint, Steinem condemned patriarchy, saying that men also have to play their roles as caregivers in raising children, otherwise the society is built on distorted human relationships.	
---	--

Voyant tool - Corpus Collocates:

The Feminine Mystique - As shown, the context is about education and career, the red words such as suffer, strange, rage, miss, emptiness, difficult, denied, dangerous, concerned, complaints, etc., describe the condition women face daily.

	Collocate	Frequency
women	problems, crisis, passive, rage, miss, abuse, unnatural, unfinished, trapped, terrible, suffered, frustrated, freedom, forbidden, complaints, suppression unhappy, violent, trap, vulnerable, reject, hostility, hatred, guilty, death, struggle, stress, scarcity, stress, poverty, misery, menace	11 9 6 5 4 3 2 1
children	guilty, fear, dehumanization, conflicts, bored	2
work	problems, career, freedom	2

Betty Friedan's 'The Feminine Mystique' - As demonstrated in the outputs obtained from Collocate Graphs, Mandala, and Document Terms, Friedan emphasized the self-image of women in American society, which can be analyzed through their education, work, and house life.

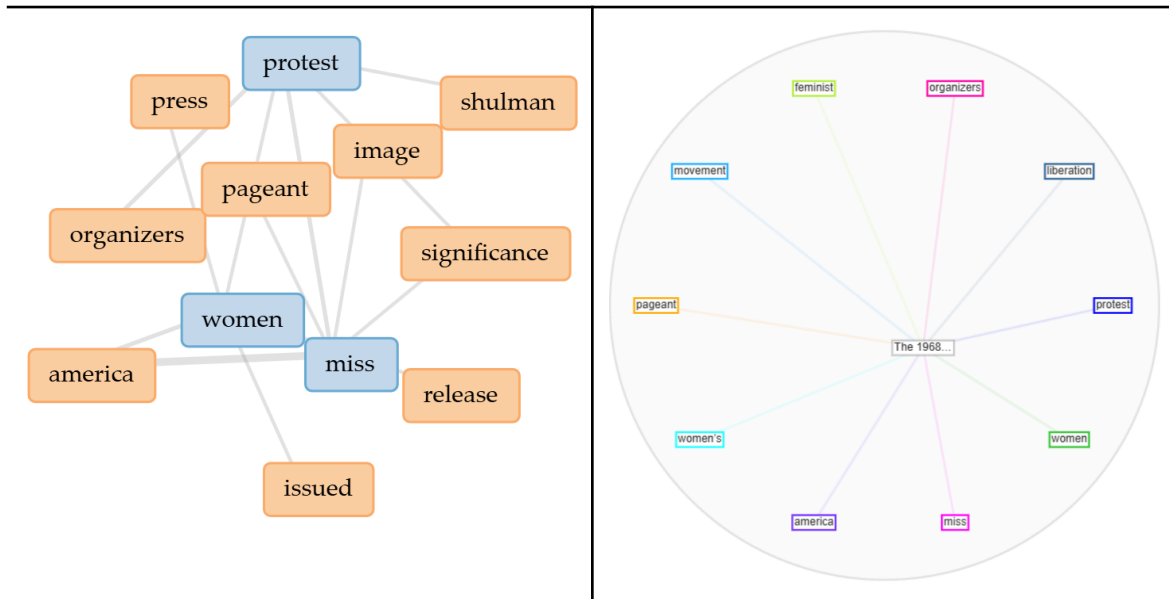


	Count	Relativity
women	1472	9,635
children	384	2,513
men	368	2,409
life	362	2,409
work	333	2,180
feminine	293	1,918
mystique	267	1,748
education	258	1,689
college	247	1,617
housewife	228	1,492
mothers	171	1,119

husband	162	1,060
society	142	929
problem	138	903
school	129	844
marriage	175	1,145
child	176	1,152

Miss America Protest - The results obtained from the aforementioned Voyant tools describe that citizens, including black and white women, protest the beauty contest since its main concept was to display women's beauty as if they were dolls, which led to the oppression of the freedom of women. By taking advantage of this protest, they also expressed their hatred towards racism and ageism, which women face very often.

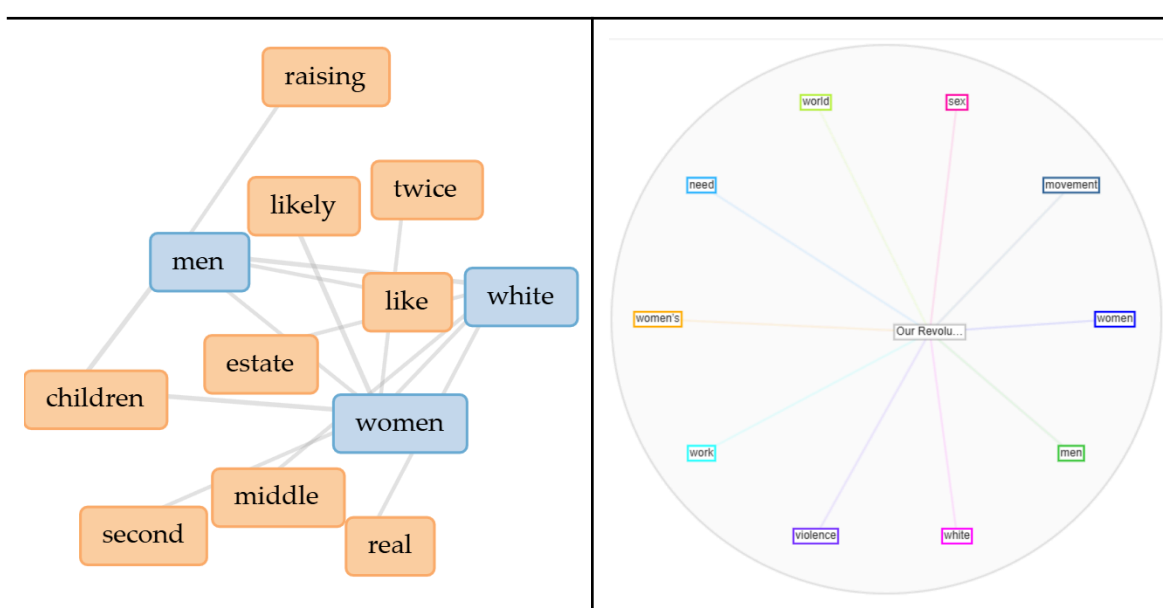
	Collocate	Frequency
protest	oppression, liberation, trash, success, freedom	1
women	oppression, freedom, beauty	1
miss	puppet, movement, protestors, feminist	1
pageant	racism	1
feminist	prominent, miss, isolated	1
organizer	boycott, ageism	1
liberation	protest, myth, consciousness	1



	Count	Relativity
protest	15	15,924
women	14	14,862
pageant, movement, feminist	6	6,369
organizers, protesters, liberation	5	5,308
racism, oppression	3	3,185
work, trash, society, role, rights	2	2,123

Gloria Steinem ‘Our Revolution Has Just Begun’ - As displayed, the word women is linked to terms such as killed, helpless, discrimination, cold, inadequate, death, complicated, and so on. The results of other tools also show that, in her speech, Steinem criticizes the inadequate treatment that women are exposed to, and they are discriminated against by society, and killed by misogynists. She draws attention to the point that society must get rid of these wrong misconceptions against women and welcome them into active duties, and provide a safe environment for their improvement in different aspects of life.

	Collocate	Frequency
violence	trafficking, females, domestic	2
women	uncomfortable, race, religions, support, myth, killed, helplessness, discrimination, equality, feminist	1
world	inadequate, death, complicated	1
movement	prison, challenging, liberation	1



	Count	Relativity
women	26	11,295
men	10	4,344
violence	9	3,910
work	8	3,475
movement	7	3,041
economic	7	3,041
children	7	3,041

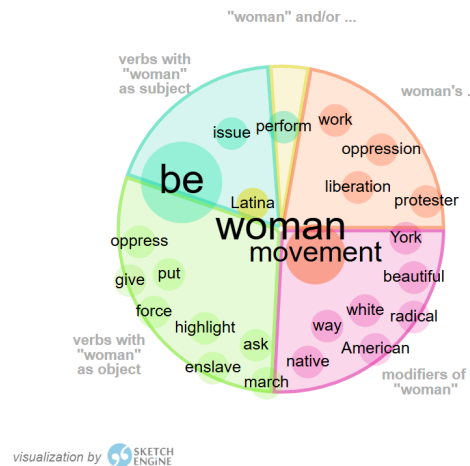
verbs with "freedom" as object	verbs with "freedom" as subject	"freedom" and/or ...
prize prizes first the freedom	move freedom moves	right rights and freedom
salute salute the wonderful freedom	have freedom has	necessity freedom and the necessity
flaunt flaunting her new freedom	be freedom has been	personhood personhood , freedom
compare freedom compared		education education , freedom
exercise exercise their human freedom	nouns modified by "freedom"	patient freedom , the patient
handle handling such freedom	right freedom , rights	will freedom and will
lose lost freedom	woman freedom , women	independence independence , the sexual freedom
win freedoms won		courage courage , freedom
want want these freedoms		sexuality sexuality , the freedom
deny denied the freedom		vision freedom , and the vision
use use their freedom		equality freedom and equality
give given complete freedom		status freedom , the status

Miss America Protest

In these results, it is shown that activists protest the enslavement of women and the oppression of. The phrases such as women's liberation movement, women's liberation protesters prove that this protest was performed for the sake of the liberation of women from these misleading stereotypes about women's roles and status in society.

modifiers of "woman"

native	Native American women
white	white women
way	ways women
beautiful	beautiful women
York	York Radical Women
American	American women
radical	Radical Women



verbs with "woman" as object

force	women are FORCED
put	put white women
ask	asked the women
enslave	women were enslaved
highlight	highlight the ways women
march	women marched
oppress	oppresses women
give	gave some women

woman's ...

movement	women's liberation movement into the
work	woman's work
protester	women's liberation protesters
oppression	women's oppression
liberation	women's liberation

verbs with "woman" as subject

be	Women Are
perform	woman performed
issue	Women issued

"woman" and/or ...

Latina	Latina and Native American women
---------------	----------------------------------

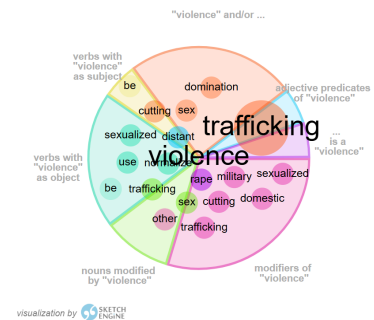
Gloria Steinem 'Our Revolution Has Just Begun' - It seems that Steinem focused on the abuse of women to draw people's attention to this subject. She condemned and demanded to take action any kind of violence committed by men against women.

modifiers of "violence"

military
military violence
sexualized
sexualized violence
trafficking
trafficking , sexualized violence
cutting
cutting , domestic violence
domestic
domestic violence
other
other violence

verbs with "violence" as object

normalize
normalizes all other violence
use
use military violence
sexualized
sexualized violence
be
is about violence



A new methodology to study the history of feminism

The Corpus of Historical American English (COHA)²⁸ - COHA helps us identify the shift from the first wave to the second wave of feminism. As shown, during the 19th century, the terms such as feminist, feminism, and businesswoman did not exist. They gradually began to be used at the beginning of the 20th century, which means that women from all classes of the society, including educated and non-educated, already realized that they need to stand up for themselves and defend their rights as humans and citizens of the country, and should be active participants in social life. While suffrage was the main theme of the 19th century, the issues regarding voting rights have always been discussed. Women just being housewives was the biggest issue during the second wave; therefore, this subject was the hot topic during this period. The term ‘freedom’ was a common term in the 19th century, compared to ‘liberation’, which emerged in the 20th century, related to the movements for civil rights, women’s rights, and so on.

Feminist:

²⁸ <https://www.english-corpora.org/coha/>

Corpus of Historical American English

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	1374	0	0	0	0	0	0	0	0	0	21	64	34	14	22	8	87	213	289	385	237
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	3.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.91	2.49	1.23	0.51	0.77	0.27	3.02	7.14	8.72	11.06	6.68
SEE ALL YEARS AT ONCE																					

Feminism:

Corpus of Historical American English

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	408	0	0	0	0	0	0	0	0	3	14	30	11	3	4	0	17	52	91	96	87
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	1.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.14	0.61	1.17	0.40	0.11	0.14	0.00	0.59	1.74	2.75	2.76	2.45
SEE ALL YEARS AT ONCE																					

Vote:

Corpus of Historical American English

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	35049	190	553	1026	768	996	1289	1762	2107	1698	1514	2208	2237	2602	2938	3089	2760	2080	1567	1704	1961
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	86.54	27.22	40.33	64.91	46.44	58.81	68.61	87.80	103.15	77.26	65.53	85.91	80.74	94.96	102.51	106.07	95.74	69.68	47.27	48.93	55.31
SEE ALL YEARS AT ONCE																					

Voting:

78

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	7523	31	66	129	121	125	234	303	358	281	221	422	411	605	693	901	547	502	379	505	689
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	18.58	4.44	4.81	8.16	7.32	7.38	12.45	15.10	17.53	12.79	9.57	16.42	14.83	22.08	24.18	30.94	18.97	16.82	11.43	14.50	19.43
SEE ALL YEARS AT ONCE																					

Housewife:

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	2145	15	30	32	28	35	45	47	40	65	92	127	137	197	277	217	263	205	96	100	97
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	5.30	2.15	2.19	2.02	1.69	2.07	2.40	2.34	1.96	2.96	3.98	4.94	4.94	7.19	9.66	7.45	9.12	6.87	2.90	2.87	2.74
SEE ALL YEARS AT ONCE																					

Suffrage:

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	4466	41	133	322	175	251	251	320	896	446	493	540	162	99	49	38	60	69	27	47	47
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	11.03	5.87	9.70	20.37	10.58	14.82	13.36	15.95	43.86	20.29	21.34	21.01	5.85	3.61	1.71	1.30	2.08	2.31	0.81	1.35	1.33
SEE ALL YEARS AT ONCE																					

Businesswoman:

Corpus of Historical American English

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	214	0	0	0	0	0	0	0	0	0	0	2	14	11	18	15	20	24	27	42	41
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	0.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.51	0.40	0.63	0.52	0.69	0.80	0.81	1.21	1.16
SEE ALL YEARS AT ONCE																					

Liberation:

Corpus of Historical American English

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	3328	14	48	41	51	38	47	26	26	38	77	79	75	335	279	370	630	487	300	200	167
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	8.22	2.01	3.50	2.59	3.08	2.24	2.50	1.30	1.27	1.73	3.33	3.07	2.71	12.23	9.73	12.70	21.85	16.31	9.05	5.74	4.71
SEE ALL YEARS AT ONCE																					

Freedom:

Corpus of Historical American English

SEARCH

CHART

CONTEXT

ACCOUNT

CLICK TO SEE CONTEXT (CLICK COLUMN HEADER FOR SECTIONS)

SECTION	ALL	1820	1830	1840	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000	2010
FREQ	37544	647	1525	1976	2062	1930	1380	1551	1552	1515	1770	1836	1682	2390	2573	2685	2075	2252	2327	1843	1973
WORDS (M)	405	7.0	13.7	15.8	16.5	16.9	18.8	20.1	20.4	22.0	23.1	25.7	27.7	27.4	28.7	29.1	28.8	29.9	33.1	34.8	35.5
PER MIL	92.70	92.67	111.22	125.01	124.70	113.95	73.45	77.29	75.98	68.93	76.61	71.44	60.70	87.23	89.77	92.20	71.98	75.44	70.20	52.93	55.65
SEE ALL YEARS AT ONCE																					

Conclusion

This thesis has reviewed the first and second waves of feminism through data from different kinds of actions, including conventions, writings, trials, meetings, and speeches. The Seneca Falls Convention was a turning point in the improvement of women's lives, aiming to provide them with their social and political rights, although no effect was observed at that time. Later, several conferences with similar concepts were held, and eventually, women were granted the right to vote after the 19th Amendment was ratified in 1920. As a result, millions of women were able to vote in elections and play a role in the political improvement of their country. When we look through the outputs of linguistic tools applied on this convention, it seems that the conference does not contain any extreme negative opinions targeted any group of people, on the contrary, ideas shared there are positively subjective opinions which were based the requirements to treat both genders equally and let them use their voting rights equally. Sentiment Analysis allows us to see the general tone of the data, while we see thematic words, clouds, and pie charts, which introduce the agenda of the conference. Voyant tool and SketchEngine conducted more linguistic analysis by showing the words either in their context or with their related words. Elizabeth Cady Stanton wrote her speech called 'Address on Women's Rights' for the conference, and here she expressed her thoughts on actual gender-related problems that distort the balance in society. In her speech, Stanton defended women's rights by highlighting that both genders are created equal in terms of intellectual and moral rights. Therefore, fundamental rights such as suffrage should be provided for women. The word cloud of the speech shows that it analyzes the statuses of men and women in society and talks about mind, will, and rights. According to the Pie Chart, 12.7% of her speech is about men's superiority, which she condemns constantly, and based on contextual phrases from WordSketch, it appears that the concept of the speech is regarding the declaration and securing of fundamental rights, including suffrage and education. While looking through the results of named entity recognition, it is possible to see male supporters such as Richard P. Hunt, who was an affluent businessperson, and James Mott, the husband of Lucretia Mott, who was one of the organizers of the conference. Martha Wright, who was the sister of Lucretia Mott, Mary Ann McClintock, Charlotte Woodward, and many others, was among the supporters of women's rights. Charlotte Woodward was born in 1830 and participated in the conference in 1840. After decades, finally she was able to see the

victory day at the end of this struggle in 1920, and she passed away in 1924.

The trial of Susan B. Anthony is a practical proof of injustice and unfair treatment that women were exposed to at that time. Susan B. Anthony was charged and fined a sum of money because she voted in the election, even though she defended her right as a citizen. The judge was also anti-feminist and a supporter of the group who were trying to keep women under their control and make them obey the male class of society. As it is seen from the word cloud of data, the trial was about the right to vote, law, and constitution, while contextual expressions from WordSketch displays that Susan B. Anthony claims that the voting right arises from citizenship, therefore it is a fundamental, personal and inalienable right that women should also have.

The second wave of feminism is one step ahead of the first wave. After the grant of the right to vote, the next goal was to secure women's roles in different fields of society, including the acquisition of the proper higher education and building a career. In her writing, she describes the miserable life of women who just did housechores and lived a monotonous life, therefore felt discontent. She expressed that women should get educated, get a paid job, and build a career; in general, they should be active members of society. However, institutional organizations, education centers, and media discouraged and manipulated them to be just housewives and mothers. With a 0.688 neutral score, it is shown that the writing does not contain the feeling of hatred or biased ideas towards any gender; on the contrary, it offers a neutral approach to solve this unpleasant situation based on subjective ideas, with a score of 0.4505 that was received during Sentiment Analysis. As appeared in the word cloud, the main subjects are children-rearing, education, and work, while according to the results of the Pie Chart, rights, freedom, and equality were the essential demands (65.8%, 17.9%, 11.9% respectively).

Miss America Protest was such a meeting that women did not just demand gender equality and equal human rights at the table, but they also expressed their frustration and decisive objection to unfair treatment. Because women required justice, the content is mainly neutral (0.687), however, the compound score is highly negative (-0.8159) because of the expressions that describe protesters' anger. Both in the word cloud and the Pie Chart, it is shown that the key goal was to win the liberation from stereotypes and standards that constrain women in a limited circle. WordSketch result of the protest data demonstrates that women were being forced, oppressed, and enslaved, therefore, this meeting was also targeted to end the suffering of women.

Gloria Steinem is a prominent public figure who is a successful businessperson who

established some organizations, such as the National Women's Political Caucus (NWPC), Women's Action Alliance, targeted to protect women's rights, to balance gender equality, and secure women's roles in political spheres. Based on the results obtained from the word cloud and the Pie Chart, in her speech, which she published on her website called Ms. Magazine, she criticized violence against women and promoted the idea of women building their careers. She also mentioned domestic violence and human trafficking targeted at women and girls. Similar to other documents of data, the sentiment of the speech is neutral, with a high proportion of positively oriented thoughts, which made the compound score high (0.9797).

The use of digital tools makes it possible to examine the context of data in different ways. Numerous methods and models in Python allow users to analyze vast amounts of data for visualization or to acquire statistical information. The use of Word Cloud and Pie Chart made it possible to detect the thematic terms of each document and how often they have been talked about. Sentiment Analysis revealed whether the emotional approach and message are positive, negative, neutral, or compound, and the features like polarity and subjectivity showed the language tone of the context and how much data is based on personal views. Named Entity Recognition (NER) allowed us to know the supporters and participants in both waves of feminism. Corpus Collocates method of Voyant tools helps to see the terms with the words that they are often used together in the text, which leads to identifying semantic patterns and connotations. WordSketch tool of SketchEngine provides users with the grammatical and collocational structure of the target term, including its uses with five categories, such as subject, object, verb, preposition, and modifier, which all reveal grammatical and syntactic information about the key term.

To conclude, this thesis analyzed the documents related to the first and second waves of feminism using six tools, in general: Sentiment Analysis, Word Cloud, NER, and Pie Chart of Python, Corpus Collocates of Voyant tool, WordSketch of SketchEngine, and COHA each of which is important to receive clear information about corpus. Feminism is an ongoing movement in the modern world. Therefore, there will be a lot of cases that need searching with different methods, including the ones that are used in this thesis.

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